

Emotional Shocks and Consumer Spending*

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Abstract

This study examines how emotional shocks affect consumer spending by using unexpected National Football League (NFL) game outcomes as a natural experiment. We analyze consumer scanner data from 83,332 households during regular seasons from 2004 to 2019, using Las Vegas pregame point spreads to identify emotional shocks. We find that unexpected losses by home teams increase household shopping trips and spending, while unexpected wins reduce spending with smaller magnitudes. These emotional impacts are concentrated within three days post-game and amplified for high-stakes games. These findings provide insights into how emotional shocks from unexpected events can influence consumer spending behavior.

Keywords: emotional state, consumer behavior, consumption

JEL Codes: D12, D91, L83

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*“The thrill of victory...
and the agony of defeat...
the human drama of athletic competition”*
— Jim McKay, ABC Wide World of Sports (1961)

1 Introduction

How might emotions influence consumers’ shopping behaviors? Consumers’ decisions are often shaped by the conditions under which they are made and emotions can play a crucial role in this process (Williams et al., 2014; Iyer et al., 2020; Sharma et al., 2023). Feelings of anxiety during crises, for example, have been linked to stockpiling and changes in food purchases (Cai et al., 2024; Hobbs, 2020; Mahajan and Tomar, 2021), while feelings of happiness and contentment are associated with increased shopping, often encouraged by sensory strategies like scents and visual displays (Mattila and Wirtz, 2001; Mohan, Sivakumaran, and Sharma, 2013). Despite these insights, a significant gap remains in understanding the causal effects of emotional shocks—unexpected events that elicit intense emotional responses—on consumer spending and the differential impacts of positive versus negative emotions on shopping behaviors.

In this paper, we leverage the unexpected outcomes of professional football games as a natural experiment to examine how emotional shocks influence consumer spending behavior. We hypothesize that unexpected negative outcomes of football games trigger negative feelings in fans, while unexpected positive outcomes elicit positive feelings. Using detailed shopping data linked to National Football League (NFL) games, we analyze whether these emotional shocks affect consumers’ shopping decisions and explore how the direction of the shock—positive or negative—differentially impacts spending behavior.

Several factors justify NFL games as a good case of a natural experiment, providing a

unique context for examining how changes in emotional states influence American fans' subsequent purchasing behaviors. First, NFL games are widely recognized as the most popular sporting events in the United States. Statistics show that the 2023 regular season attracted an average of 17.9 million viewers per game across all networks.¹ Therefore, the emotional reactions tied to NFL games reflect an intense, common, and recurring emotional experience for individuals. Second, NFL games have highly dedicated local fan bases compared to other sports events ([World Sports Network, 2022a,b](#)), capturing a significant share of the local TV audience. This strong fan attachment to local teams facilitates support identification. Third, the extended game season, structured competition, detailed statistics, and well-organized betting market enhance the ability to identify effects and explore mechanisms. Substantial variations in team performance across seasons also provide an ideal setting to investigate the effects of football events.² Lastly, the unexpected outcomes of football games have been used to measure emotional shocks in well-established studies, such as [Card and Dahl \(2011\)](#) and [Eren and Mocan \(2018\)](#).

In our analysis, an emotional shock occurred if the outcome of a game deviated from the fans' rational expectations of a game outcome, as measured by the Las Vegas pregame point spread. Outcomes that differ from the pregame point spread (i.e., unexpected outcomes) serve as either positive or negative emotional shocks for fans. We categorize such emotional shocks into three types: upset losses (i.e., losses by a team when it was expected to win), close losses (i.e., losses by the team when the outcome was uncertain beforehand), and upset wins (i.e., wins by a team when it was expected to lose). To the extent that the pregame point spread efficiently predicts game outcomes, controlling for it allows us to interpret any differential impact between a win and a loss as the causal effect of the game outcome.

To investigate household purchase decisions, we utilize the NielsenIQ Homescan

¹Source: [NFL News](#), Jan 10, 2024. According to Nielsen's annual ranking of the 100 most-watched programs in 2023, NFL Games accounted for 93% of the most-watched TV programs and NFL games dominate weekly television ratings each fall.

²There are 14 different teams that have won the NFL championship in the past twenty years.

Panel, which provides comprehensive information on households and their buying behaviors. This dataset features a 19-year rotating panel of households, recording nearly all their purchases from major retailers and grocery stores. It includes detailed demographic data about households, as well as extensive information on products and purchases, such as prices, quantities, promotional activities, and a well-structured product hierarchy. Focusing on households with an exclusively local NFL team, we narrow our dataset to purchase records from 83,332 unique households during NFL regular seasons from 2004 to 2019.

Using the Homescan dataset linked to NFL team records, we examine how emotional shocks from NFL games influence household spending, categorizing expenditures into three groups: food, non-food, and alcohol. Our analysis focuses on Sunday regular-season games, which comprise approximately 86% of all regular-season games during the study period. We concentrate on home division games, which are both highly emotionally charged for local audiences and typically spaced apart in the schedule. This spacing minimizes the likelihood of overlapping exposure periods that could otherwise distort the relationship between game timing and consumer purchase behavior.³ Conditioning on the Las Vegas pregame point spread, we regress the natural logarithm of spending on emotional shock variables, controlling for household characteristics and holiday dummies. We also include individual, week, and season-fixed effects in all specifications, which enables us to identify the causal effects of emotional shock exposure through random and exogenous variation in game outcomes.

The main findings of this study indicate that unexpected home team losses lead to significant increases in both total spending and shopping trips, with spending rising across all categories and showing the greatest impact on food items. In contrast, unexpected wins lead to a reduction in spending. Losses in closely contested games show no significant effect on household spending. These changes in spending behavior are

³Results for all games and home games are reported in [Appendix B](#). These findings are consistent with the main results, albeit with smaller magnitudes.

concentrated within a narrow window of 1–3 days.

We extend our analysis in several ways. First, we analyze spending behavior by game type, focusing on the salience of each game. Our findings indicate that the effects of upset outcomes are particularly pronounced after high-stakes games, highlighting the importance of game intensity in driving household spending responses. Second, we investigate the heterogeneous effects of emotional shocks across demographic groups (i.e., age, race, and income). We find that upset losses have a stronger impact on spending among households with older heads and higher disposable incomes. Meanwhile, upset wins show more pronounced effects on spending among households with non-white heads. Third, we disaggregate food spending into perishable and non-perishable categories to examine potential substitution effects between food-at-home and food-away-from-home consumption. Results show that spending for non-perishable food items experienced larger changes, suggesting that consumers may prioritize stocking durable goods following emotional shocks.

The findings remain consistent across several robustness checks. First, we employ alternative methods including different cutoffs for the point spread and using different parameterizations to define emotional shock. Second, we test alternative model specifications by varying fixed effects, control variables, and clustering levels to address potential concerns. Third, we analyze a restricted sample of counties with stronger attachment to the local team, further validating the results. Additionally, a placebo test confirms that household total spending is unaffected by home games scheduled for the following month.

Our findings connect with several strands of literature. The first strand delves into consumers' emotional shopping behaviors. A significant portion of this literature documents impulse buying, defined as a sudden and compelling urge to purchase goods immediately ([Rook, 1987](#); [Beatty and Ferrell, 1998](#)). Impulse buying is shaped by internal factors such as personality traits and motives, as well as external stimuli like marketing

strategies, with emotions playing a central mediating role (Redine et al., 2023; Iyer et al., 2020). While 80% of impulse purchases occur in physical grocery stores (Saleh, 2023), recent research has increasingly focused on online and social commerce platforms (Djafarova and Bowes, 2021; Huang, 2016; Sundström, Hjelm-Lidholm, and Radon, 2019; Yang et al., 2021), leaving impulsive behaviors in traditional grocery environments relatively underexplored (Lord, Putrevu, and Olson, 2023). Moreover, existing studies provide mixed evidence on whether impulse purchases are primarily driven by positive emotions, negative emotions, or both (Beatty and Ferrell, 1998; Mortimer, Bougoure, and Fazal-E-Hasan, 2015; Verhagen and van Dolen, 2011; Vohs and Faber, 2007), often relying on survey data with small sample sizes and self-reported emotional states, limiting causal insights.

Another strand of literature focuses on the impact of emotional states on consumption, particularly food choices under negative emotions. Stress and negative emotions have been shown to increase snacking and unhealthy eating (Gillebaart et al., 2022; Oliver and Wardle, 1999; Schultchen et al., 2019; Sproesser, Schupp, and Renner, 2014), with Cornil and Chandon (2013) linking fans' unhealthy eating habits following team losses to self-affirmation processes. More recently, broader shifts in consumption patterns driven by negative emotions have been documented during the COVID-19 pandemic (Gligorić et al., 2022; Cai et al., 2024).

We address gaps in these strands of literature by leveraging a comprehensive household scanner dataset that captures detailed in-store purchase records from a representative sample of consumers. This approach allows us to analyze how emotional shocks influence both the frequency and magnitude of spending across product categories, offering robust, real-world insights. By simultaneously measuring the effects of both negative and positive emotional shocks, we provide insights into their distinct impacts. Furthermore, the dataset's granular tracking of purchases enables us to directly link emotional changes to observable shopping behaviors, enhancing our understanding of

how emotions drive consumer decisions in practical, high-stakes contexts.

Our study also intersects with research on the broader behavioral impacts of emotional shocks from unexpected sports outcomes. For instance, [Card and Dahl \(2011\)](#) and [Cardazzi et al. \(2022\)](#) show that heightened emotional responses to home football and basketball games significantly increase domestic violence rates in the host city. In contrast, [Ivandić et al. \(2024\)](#) find no significant effect of unexpected football results involving Manchester United and Manchester City on domestic abuse. Emotional shocks from sports outcomes also influence financial and academic outcomes. [Edmans, García, and Norli \(2007\)](#) report short-lived stock market declines after national soccer team losses, attributing these drops to changes in investor mood rather than economic factors. Similarly, [Lindo, Swensen, and Waddell \(2012\)](#) find that male students' GPAs decline during football seasons due to increased alcohol consumption and partying. Beyond these domains, sports-related emotional shocks have been shown to influence judicial sentencing ([Eren and Mocan, 2018](#)), asylum decisions ([Chen and Spamann, 2014](#)), and electoral outcomes ([Healy, Malhotra, and Mo, 2010](#)). Expanding this body of work, our findings document how emotional shocks from NFL games influence consumer spending behavior.

The remainder of this paper is organized as follows. Section 2 presents the data used in our analysis along with relevant summary statistics. Section 3 describes the econometric methodology. Section 4 presents the results. Finally, Section 5 provides a summary and concluding thoughts.

2 Data and Sample Construction

2.1 *NielsenIQ Homescan Data*

We use the NielsenIQ Homescan Panel covering 2004–2019 to measure household grocery purchases.⁴ The dataset consists of a representative panel of households in the U.S., ranging from approximately 40,000 to 60,000 per year. These households use in-home scanners to record the Universal Product Codes (UPCs) of all consumer packaged goods they purchase from any retail outlet.⁵ The Homescan data provides detailed information for each item purchased on each shopping trip, including transaction metadata such as purchase date, store/retailer information, and panelist identifier. Additionally, the Homescan Panel offers granular details on product and transaction details, including transaction price, amounts of quantities purchased, deal/sale usage, the total expenditure on the item, and detailed product identifiers.

NielsenIQ categorizes millions of UPCs into ten departments (i.e., the broadest category), six of which are food-related: dairy, deli, dry grocery, fresh produce, frozen food, and packaged meat. For our main analyses, we combine these six departments into a single "Food" category, representing approximately 65% of the total expenditure tracked by the Homescan Panel. The remaining departments are grouped as follows: general merchandise, health and beauty, and non-food grocery are combined into a "Non-food" category, while alcohol is classified as a separate, individual department.

NielsenIQ recruits panelists using direct mail and Internet advertising and works to ensure the sample represents a broad demographic spectrum. On average, households remain in the panel for about 54 months.⁶ Homescan households annually report

⁴These data are available for academic research through a partnership with the Kilts Center at the University of Chicago Booth School of Business. Visit <http://research.chicagobooth.edu/nielsen> for more details.

⁵The data include purchases from supermarkets, convenience stores, mass merchandisers, club stores, drug stores, and other retail channels.

⁶For a detailed description of the procedures for recruiting and retaining panelists, please refer to [Muth, Siegel, and Zhen \(2007\)](#) and the Consumer Panel Dataset Manual. This dataset features projection

a wealth of demographic and socioeconomic characteristics, including age, education, marital status, race, household size, and income.⁷ These attributes are categorized into different groups within the dataset. For our study, we reclassify some of these categories to serve as control variables.⁸ For households with two heads, we use their average age and education levels.

2.2 *Linking Homescan Panel to NFL Team Records*

We link the Homescan Panel to National Football League team records for local franchises. To identify local franchises, we follow the approach of [Card and Dahl \(2011\)](#). Specifically, we focus on states and cities with NFL teams. For states with only one NFL team, all panelists within that state are linked to that one NFL team. In states with multiple NFL teams located in different cities, panelists are assigned based on their city of residence to the corresponding local team. We exclude the New York Jets and New York Giants from our analysis due to their shared home city, as well as the Los Angeles Chargers and Los Angeles Rams due to their recent relocation.⁹ Additionally, we exclude panelists who had ever moved to other cities without the same home team. [Figure 1](#) illustrates the distribution of local NFL team franchises.

The NFL season begins with the regular season, which spans 17 weeks from early September to late December or early January. This is followed by the playoff season, which lasts about 4 weeks.¹⁰ Due to the annual rotation of panelists and the inconsistent

weights designed to ensure the sample is representative of the overall U.S. population. These weights enable researchers to extrapolate data to national, regional, and scantrack market levels. [Bronnenberg et al. \(2015\)](#) provide a summary supporting the representativeness of the panel of U.S. consumers.

⁷All demographics data is collected during the fall of the year prior to the panel year, except for household income data, which is for two years prior to the panel year.

⁸Age, education, and income information in NielsenIQ’s consumer data are classified into 10, 7, 20 groups, respectively. Following [Allcott et al. \(2019\)](#), we assign the mean value of each group for age and income, and assign a representative number of years for each education group. For instance, an individual who has completed college is assigned 16 years of education.

⁹The Chargers moved from San Diego to Los Angeles in the 2017 season, and the Rams moved from St. Louis to Los Angeles in the 2016 season.

¹⁰From the 2004 to 2019 seasons, there were a total of 7,893 NFL games, of which 95.88% (7,568) were regular-season games.

coverage of the final week of the regular season in the Homescan panel year, our analysis is restricted to the first 16 weeks of regular-season games, which account for 86% of all regular-season games. To simplify our empirical design, we focus specifically on regular-season Sunday games. In addition, we restrict our analysis to home division games, which are both highly relevant to local audiences and typically spaced apart in the schedule.^{11,12} By focusing on these matchups, we reduce the chance of overlapping exposure periods that could distort the relationship between game timing and consumer purchase behavior. As [Table 1](#) shows, there is a total of 987 games included in our sample. Having imposed these restrictions, our final sample includes 83,332 unique households matched to 30 NFL teams during the 2004–2019 period.¹³

Linking the Homescan panel data with the records of local NFL teams allows us to examine how consumer spending behaviors change during the week following a Sunday NFL game throughout the regular season. [Table 2](#) offers a comprehensive summary of household expenditures and shopping trips and [Table 3](#) reports households’ demographic characteristics. We deflate expenditures and incomes to real 2012 dollars using the consumer price index for urban consumers for all items. Particularly, in [Table 2](#), we present a descriptive summary of household expenditures across different categories on game day and compare them with spending patterns in the first two days afterward (Monday and Tuesday) and the subsequent week (Monday through Saturday). Additionally, we compare the average expenditure and the number of trips during game weeks with those during bye weeks when no game was played. On average, households in our sample spend approximately \$104 each week. The majority of this expenditure, over 60% (about \$66), is on food, with only 3.5% (about \$4) allocated to alcohol.¹⁴ This spending

¹¹[Table A1](#) presents NFL football teams, organized by divisions within the American Football Conference (AFC) and the National Football Conference (NFC).

¹²Of the total 987 home division games during the 2004–2019 regular seasons, only 60 games (around 6%) are scheduled back-to-back.

¹³Detailed observations for each NFL team over the years are presented in [Table A2](#) in the appendix.

¹⁴According to the U.S. Bureau of Labor Statistics (BLS), the average annual expenditure on food at home for all consumer units (with an average of 2.5 persons per unit) is \$3,921. This figure is slightly higher than our calculation of \$3,432, which can be partially attributed to our focus solely on packaged

pattern is consistent across different time windows. During game weeks, households tend to spend more on alcohol and non-food items compared to during bye weeks, though their food expenditure is slightly reduced.

2.3 *NFL Teams' Predicted and Actual Outcomes*

Sports spread betting involves predicting the outcome of a match based on a point spread, which indicates the expected score difference between two teams. In NFL games, betting on game outcomes is organized through Las Vegas bookmakers. For example, if Team A is favored by 3 points over Team B, Team A must win by more than 3 points for a bet on Team A to be successful. The closing value of the point spread captures the market's prediction of the game's outcome.

[Card and Dahl \(2011\)](#) and [Eren and Mocan \(2018\)](#) demonstrate that pregame point spreads are effective predictors of NFL game outcomes. Building on their findings, we gather data on pregame point spreads and final scores of all NFL games played during the 2004-2019 seasons¹⁵ and present the relationship between the actual and predicted point spread in each game in [Figure 2](#). A simple regression of the actual spreads on the predicted spreads yields a coefficient of 1.05 (standard error of 0.03) with a high R^2 value of 0.18. These findings suggest that the predicted point spreads are reliable predictors of NFL game outcomes.¹⁶

With the reliability of the point spreads established, we then categorize them into three distinct groups. Following the approach of [Card and Dahl \(2011\)](#) and [Eren and Mocan \(2018\)](#), we classify the games into three ex-ante categories, as shown in [Figure 2](#): (i) predicted win if the point spread is -4 or less, (ii) predicted close if the point spread is between -4 and 4 , and (iii) predicted loss if the point spread is 4 or more. We

foods. Visit [BLS Consumer Expenditures \(Annual\) News Release](#) for more information.

¹⁵The Spreadspoke company offers comprehensive data on NFL scores and betting information. The dataset can be accessed through Kaggle at the following link: [NFL Scores and Betting Data](#).

¹⁶Our estimated effect closely matches the results reported by [Card and Dahl \(2011\)](#) and [Eren and Mocan \(2018\)](#).

use different point spread cutoffs to assess the robustness of our results, which will be detailed in [Section 4.3A](#).

Panel A and Panel B of [Table 1](#) present a summary of the actual and predicted outcomes of the 987 regular-season Sunday home division games included in our sample. Of these games, 39.51% were predicted as wins, 15.50% as losses, and 44.98% as close matches. The teams lost approximately 25.38% of the games they were predicted to win by four or more points (i.e., upset losses) and won about 19.61% of the games they were predicted to lose by the same margin (i.e., upset wins). For games predicted to be close, approximately 50% ended in losses (i.e., close losses).

[Table 1](#) also highlights important features of NFL games that are explored in subsequent analyses. In Panel C, we present the starting times of home division games. A large share of these games (91%) started at or before 7 PM, indicating that only a small number of games were played at night. In Panel D, we consider the emotional salience of a game: whether the game had an unusually high number of sacks, turnovers, or penalty yards.¹⁷ Approximately 55% of games featured a high number of sacks, turnovers, or penalty yards.

3 Empirical Methodology

To estimate the impact of emotional shocks from unexpected NFL game outcomes on consumer spending, we estimate regressions linking household spending to shock exposure variables. We employ the following specification:

$$\begin{aligned}
Ln(Spending_{ids}) = & \lambda_0 + \lambda_1 Upset Loss_{ks} + \lambda_2 Close Loss_{ks} + \lambda_3 Upset Win_{ks} \\
& + \lambda_4 Predicted Win_{ks} + \lambda_5 Predicted Close_{ks} + \lambda_6 Predicted Loss_{ks} \\
& + \mathbf{X}'_{iks}\beta + \mu_i + \gamma_k + \delta_s + \epsilon_{ids}
\end{aligned} \tag{1}$$

¹⁷We classify games as salient if they feature frustrating occurrences such as five or more sacks, five or more turnovers, or 100 or more penalty yards.

where $Spending_{idks}$ is the amount of dollars spent on goods at home by household i on day d of week k in game season s .¹⁸ We focus on household spending on three time windows: game day (Sunday), the following two days (Monday and Tuesday), and the post-game week (Monday through Saturday). The variables *Predicted Win*, *Predicted Close*, and *Predicted Loss* are mutually exclusive indicators indicating the expected outcomes of a game based on the point spread in the betting market. These indicators represent whether the team is predicted to win, have a close game, or lose, respectively. The variables *Upset Loss*, *Close Loss*, and *Upset Win* are interactions between these predicted outcome dummies and the actual game results. Specifically, *Upset Loss* indicates that the team was predicted to win but ended up losing, while *Upset Win* indicates the opposite, where a loss was predicted, but the team won. *Close Loss* denotes a close game was predicted, but the team lost.¹⁹ The variable $\mathbf{X}'_{iks}$ includes various observed household characteristics, including age, years of education, indicator of children, household size, marital status, race, income, and internet access. It also considers household spending prior to the games to control for potential stockpiling behavior and includes a holiday indicator to account for the changes in spending during national holidays. The term μ_i captures the household fixed effects; γ_k and δ_s denote week and season effects, respectively; and ϵ_{iks} is the error term in the model. Standard errors are clustered at the household level. Our results remain robust even when we cluster at the season $\times$ week level or use two-way clustering (i.e., at the household and season), as detailed in Section 4.

In this model, weeks without a game (i.e., bye weeks) serve as the reference category.²⁰ The key coefficients of interest are λ_1 , λ_2 , and λ_3 , which capture the effects of an

¹⁸We apply the inverse hyperbolic sine (IHS) transformation to this variable to account for zero values in spending.

¹⁹In practice, we create the *Upset Loss*, *Close Loss*, and *Upset Win* as follows: $Upset Loss = \mathbb{1}(\text{Predicted Win}) \times \mathbb{1}(\text{Loss})$, $Upset Win = \mathbb{1}(\text{Predicted Loss}) \times \mathbb{1}(\text{Win})$, $Close Loss = \mathbb{1}(\text{Predicted Close}) \times \mathbb{1}(\text{Loss})$.

²⁰This approach is consistent with methodologies adopted in related studies (see, e.g., [Card and Dahl, 2011](#); [Eren and Mocan, 2018](#); [Ivandić et al., 2024](#)).

upset loss, a close loss, and an upset win on consumer spending, respectively. These coefficients represent average effects across all households in our sample—including those likely unaffected by NFL games.²¹ Positive values of these coefficients would imply that unexpected game outcomes lead to increased consumer spending, whereas negative values would suggest the opposite effect.

Our identification strategy relies on the key assumption that NFL game outcomes are as good as random after controlling for the pre-game point spread. Betting markets efficiently aggregate all publicly available information that could predict game outcomes. When we control for the point spread in Equation (1), any remaining variation in game outcomes represents unexpected shocks, allowing us to estimate the causal effect of emotional responses to these surprises.²²

4 Results

In this section, we present our key findings on how emotional shocks impact household spending across various categories, both immediately and over time. We then extend our analysis to examine the heterogeneity in the effects of emotional shocks on spending decisions. Finally, we discuss several robustness checks using alternative specifications and sample selections to ensure the reliability of our results, followed by the results of falsification tests.

²¹Because these estimates include households that may not be emotionally affected by NFL outcomes, our results likely represent a lower bound of the true impact for affected households.

²²Mathematically, for a binary random variable y representing the actual outcome of a game, with mean probability p , the expectation $E[y|p, Z] = E[y|p]$ holds true for any set of factors Z . In other words, when conditioned on the mean p , the actual outcome y is independent of Z . If the Las Vegas spread can effectively predict NFL game outcomes, we can express the mean probability p as a function of the point spread, $p(S)$. Assuming that $p(S)$ is invertible and independent of Z , we have $E[y|S, Z] = E[y|p, Z] = E[y|p]$. This indicates that the actual outcome y is independent of Z when conditioned on the point spread S (i.e., predicted outcomes).

4.1 Main Results

Table 4 presents our main results from estimating Equation (1), examining total household spending across three time windows: game day (Sunday), the following two days (Monday-Tuesday), and the postgame week (Monday-Saturday), shown in columns (1), (2), and (3), respectively.²³ The results in column (1) show that an upset loss by the home team during a home division game leads to an immediate, statistically significant increase of 2.3% in household spending on Sunday. In contrast, a close loss yields a positive but statistically insignificant change in spending. An upset win, meanwhile, is estimated to reduce household spending by 1.3%, though this effect is not statistically significant.²⁴

Second, although Homescan records the date for each purchase, it does not specify the exact time of the transaction. This limitation affects our ability to determine whether a purchase was made before or after a game on the game day. To mitigate this issue, we assess whether the effects on consumer spending vary depending on the timing of the game within the day.

Households' spending responses to emotional shocks become more pronounced over the next two days (Monday-Tuesday). As shown in column (2), spending increases significantly by 5.2% following upset losses, more than doubling the effect observed on Sunday. Conversely, an upset win leads to a 4.1% reduction in spending, a significant change compared to Sunday, where the effect was insignificant. Close losses continue to show no statistically significant impact. Over the longer window (Monday-Saturday) in column (3), the effects of emotional shocks largely diminish, with no significant changes

²³Complete estimates of all coefficients, including control variables, are reported in Appendix Table A3.

²⁴Although Homescan records the date of each purchase, it does not capture the exact time of the transaction, which limits our ability to determine whether purchases on game day occurred before or after the game. To address this limitation, we evaluate whether consumer spending effects vary based on the timing of the game within the day. Specifically, we separately analyze games with early kickoffs (1 PM or earlier) and late kickoffs (7 PM or earlier). The results, presented in Appendix Table A5, indicate that the effects are similar across both time slots, suggesting that our findings are robust to variations in game timing.

in spending following either upset losses or wins. These findings suggest that emotional shocks exert their strongest influence on consumer spending within the initial days after the event, fading by the end of the week.

To further examine whether households respond to emotional shocks by adjusting their shopping frequency rather than merely altering their spending levels, we modify Equation (1) by replacing the dependent variable with the number of grocery store visits.²⁵ Given the incident-based nature of trip data, we employ a Poisson model for the estimation. The results, presented in columns (4), (5), and (6) of Table 4, show that upset losses increase household shopping trips by 1.3% on game day and by 3% over the following two days. In contrast, upset wins reduce trips by 3.6% during the first two days following the game, while close losses have no statistically significant effect. Examining the full Monday-Saturday window, we find no persistent effects on shopping trips from either upset losses or wins. The parallel patterns in shopping frequency and spending suggest that emotional shocks influence both how often households shop and how much they spend.

We next analyze how emotional shocks affect spending across specific categories. Figure 3 presents the estimated effects for food, alcohol, and non-food items over our three time windows: Sunday (blue circles), Monday to Tuesday (green triangles), and Monday to Saturday (orange squares), with bars in the figure representing 95% confidence intervals. The strongest effects appear for food spending following upset losses, with a 4.33% increase during Monday-Tuesday and a 1.70% increase on Sunday. Non-food spending shows a similar pattern, increasing by 2.16% during Monday-Tuesday and 1.70% on Sunday.²⁶ Alcohol spending shows a modest increase of 0.84% following upset losses, but unlike other categories, this effect persists throughout the Monday-Saturday period.

²⁵Shopping trips are defined by the frequency of visits made to purchase groceries. Each visit is counted as one trip, regardless of whether it involves stops at multiple stores.

²⁶While the NielsenIQ Homescan Data does not cover data on food purchased away from home, the observed impact on non-food purchases, which have fewer substitutes outside the home, provides further evidence that consumers' shopping decisions are influenced by emotional shocks.

In contrast to the effects of upset losses, upset wins lead to significant reductions in spending across food and non-food categories. Food spending drops by 3.88% during Monday-Tuesday and 2.67% on Sunday, while non-food spending declines by 3.33% during Monday-Tuesday. Alcohol spending shows no statistically significant changes after upset wins, suggesting that positive emotional shocks do not affect alcohol spending patterns. Close losses show negligible effects across all categories and timeframes, with small and statistically insignificant changes. These results indicate that emotional shocks from unexpected game outcomes primarily affect food and non-food spending during the first few days after games.

Overall, our main findings reveal four important insights. First, unexpected losses in emotionally charged games significantly increase consumer spending, both on the game day and in the short period that follows, while close losses have a weaker and statistically insignificant effect. Second, unexpected wins reduce consumer spending, though these effects emerge after the game day and are more modest than those following an upset loss. Third, the changes in consumption spending can be partially attributed to changes in shopping trips. Finally, examining spending by category shows that food expenditures are most sensitive to emotional shocks, with alcohol and non-food purchases also affected, albeit to a lesser extent.²⁷

4.2 *Extension Analyses*

A. Emotional Salience Games

Having established the baseline effects of unexpected game outcomes on household spending, we next examine whether these effects vary with game intensity. Our hypothesis is that more emotionally charged games should produce stronger spending responses. To identify emotionally salient games, we focus on three frustrating events

²⁷The main results for all games and home games are presented in Appendix [Table B3](#) to [Table B5](#). The findings reveal similar patterns to the main analysis, although the effects are smaller in magnitude.

that could heighten emotional engagement: five or more sacks, five or more turnovers, or 100 or more penalty yards.²⁸ These events occur in roughly half of our sample games.

Table 5 presents the effects of emotional shocks on household spending, comparing more emotionally salient games (columns 1–3) to less salient games (columns 4–6). Following upset losses in salient games, spending increases by 2.0% on Sunday and 9.2% during Monday-Tuesday, with effects persisting at 2.9% through the following week. For less salient upset losses, the increases are more modest: 1.8% on Sunday and 4% during Monday-Tuesday, with no statistically significant effect over the week.

The importance of game intensity extends beyond upset losses. Close losses in salient games generate significant spending increases of 2.4% during Monday-Tuesday and 1.7% during the following week, while similar losses in less salient games show no significant effects. Upset wins display the largest contrast: in salient games, spending decreases sharply by 7.3% on Sunday and 11.5% during Monday-Tuesday, whereas less salient upset wins produce no significant changes. These systematic differences between salient and non-salient games suggest that emotional intensity substantially amplifies consumer responses to unexpected outcomes.

B. Types of Households

We further explore whether demographic characteristics influence households' responses to unexpected game outcomes. Table 6 presents these results by income level (below average vs. above average), age (under 65 versus 65 and older), and race (non-white versus white).

The income-level analysis, presented in columns 1-3, reveals perhaps the most intriguing patterns, showing differences in both the magnitude and timing of responses. Following upset losses, below-average income households response more quickly with

²⁸In the study by Card and Dahl (2011), a game is defined as frustrating if it involves four or more sacks, four or more turnovers, or 80 or more penalty yards, which collectively account for 40% of all games in their sample.

a 2.7% spending increase on Sunday, followed by a moderate 3.7% increase during Monday-Tuesday. In contrast, above-average income households show a delayed but substantially stronger response, with no significant change on Sunday but a 7.4% increase during Monday-Tuesday. The pattern of responses to upset wins also differs meaningfully between income groups: below-average income households show significant spending reductions of 4.8% during Monday-Tuesday, while above-average income households demonstrate no significant changes in their spending patterns. These income-based differences suggest that lower-income households respond more quickly and show consistent reactions to both positive and negative shocks, while higher-income households display delayed but larger responses to negative shocks only.

The age-based analysis, presented in columns 4-6, shows distinct temporal patterns in spending responses. Following upset losses, younger households (under 65) demonstrate immediate reactions—spending increases by 2.6% on Sunday and rises further to 3.8% during Monday-Tuesday. Older households (65+) display a markedly different pattern: while showing no statistically significant spending changes on Sunday, they exhibit a substantial 9.0% increase during Monday-Tuesday. This delayed but stronger response among older households suggests potential differences in how age groups process and react to emotional shocks. However, the age groups show no statistically significant differences in their responses to upset wins, with both groups reducing spending by similar magnitudes (3.6% for younger and 4.1% for older households during Monday-Tuesday). Neither age group shows significant responses to close losses, and importantly, neither exhibits persistent effects through the week following unexpected outcomes.

Columns 7–9 present the results examining racial differences in spending responses, which prove generally modest. While non-white households show larger point estimates for spending changes—particularly a 5.6% increase during Monday-Tuesday after upset losses and a 5.8% decrease after upset wins—formal difference tests indicate these variations from white households’ responses are not statistically significant. Both racial

groups show similar patterns in terms of the timing and direction of effects, with the strongest responses consistently appearing during the Monday-Tuesday period following games.²⁹

C. Spending across Different Food Categories

Our main results show changes in food spending following unexpected game outcomes. However, these findings are limited to food purchased for home consumption, as the Homescan Panel data does not capture restaurant spending. To better understand the nature of these spending changes, we examine perishable and non-perishable food categories separately. If our results merely reflect substitution between home and restaurant meals, we might expect stronger effects for perishable items, which are more similar to restaurant meals.

[Table 7](#) presents the relative effects of emotional outcomes on household food spending, with the first three columns focusing on perishable food and the last three columns on non-perishable food.³⁰ We find that an upset loss leads to an increase in consumer spending on both perishable and non-perishable foods on the game day and for the two days following. However, the increase in spending on perishable foods is smaller compared to non-perishable foods. We also find that a close loss has a minimal positive impact on spending for both food categories. Conversely, an upset win results in a decrease in consumer spending on both perishable and non-perishable foods after the game, with a notably larger reduction in spending on non-perishable foods. The findings suggest that, to some extent, a substitution exists between spending on food at home and dining out. Nevertheless, emotional shocks still have a significant impact on perishable food expenditures at home.

²⁹The results of the extended analysis, covering both all games and home games, are presented in Appendix [Table B6](#) and [Table B7](#). The findings align with the main analysis, though the effects are generally smaller in magnitude.

³⁰Perishable food includes dairy, deli, fresh produce, and packaged meat departments. Nonperishable food includes dry grocery and frozen foods departments.

4.3 Robustness of Spending Results

We conducted several robustness checks to ensure the validity of our results. These included testing alternative parameterizations of Equation (1), exploring different model specifications, applying alternative sample selection and data cleaning methods, and analyzing a more restricted sample of households with stronger attachments to their local team.

A. Alternative parameterization

In our main analysis, we define unexpected game outcomes using a discrete parameterization with cutoff values of -4 and 4. To assess the robustness of our results, we first retain the discrete parameterization and test alternative cutoff values of -3 and 3. The results, presented in Table A6, confirm that the impacts of emotional shocks on household spending remain consistent with the main specification. Next, we explore an alternative approach to the discrete parameterization used in Equation (1), drawing on the methods of Card and Dahl (2011) and Eren and Mocan (2018). This approach incorporates a cubic polynomial of the point spread and interacts it with an indicator for home team losses. Figure 4 visualizes the estimated interaction effect across a range of point spreads, along with the corresponding 95% pointwise confidence intervals. The results indicate that the effect of a loss on household total spending decreases as point spreads increase and remains significantly different from zero when the pregame point spread is less than zero.

B. Alternative model specification

To further assess the robustness of our results, we replace season and week-of-season fixed effects with week-by-season fixed effects, enabling more granular control of unobservable factors. The results, presented in Table A7, confirm the consistent impact of upset losses and close losses from NFL games on household spending, although no

significant effects are observed for upset wins. Additionally, we modify our control for pre-game household spending. Instead of controlling solely for spending on Saturday (the day before the game), we extend the control period to include household spending across the entire week preceding the game. This approach evaluates whether our results are sensitive to the definition of baseline consumption patterns. The results, reported in [Table A8](#), remain consistent with our main analysis, suggesting that precautionary inventory stocking before game day does not confound the observed effects of emotional shocks on household spending. Furthermore, [Table A9](#) presents baseline estimates adjusted for alternative clustering methods. While our main analysis clusters standard errors at the household level, we also replicate the results by clustering standard errors at both the household and game-season levels to examine whether clustering affects the estimated impact of emotional shocks. The results remain nearly identical, indicating that clustering does not present a significant issue in our analysis.

C. Addressing potential data issues

To ensure data quality, NielsenIQ implements a purchasing threshold over a 12-month period to exclude households that report only a minimal portion of their expenditures. Despite this, we observe that approximately 10% of households report zero total spending for more than a month. To address this issue, we exclude these households and reestimate our main specification. The results, presented in [Table A10](#), align with our main findings. In addition, our main analysis excludes data from the final week of the regular season due to the annual rotation of panelists and inconsistent coverage in the Homescan panel during this period. As a robustness check, we incorporate household spending data from the final week of the regular season. For long-term panelists, we use their continuous spending records, including the final week. For panelists with only one year of data, we estimate their final-week spending based on their average daily spending over the year. The results, reported in [Table A11](#), remain consistent with the

main findings, further reinforcing the robustness of our analysis.

D. A restricted sample with a stronger attachment

In our main analyses, we identify local NFL franchises by focusing on states with a single exclusive NFL team and home cities located in states with multiple NFL teams. Panelists residing in these states or cities are linked to their respective local teams. However, recognizing that not all residents in states with an exclusive NFL team (e.g., Illinois) exhibit strong attachment to their state team, we refine our sample to focus on areas with a higher degree of local team loyalty. To define fandom at the county level, we employ multiple criteria: the location of the home stadium, geographic distribution of Twitter followers³¹, and ticket sales data³². Specifically, we include panelists residing in counties with home stadiums, as well as in neighboring counties characterized by a high concentration of local team followers on Twitter and substantial ticket sales for the team. The results, presented in [Table A12](#), reveal a similar pattern of emotional shocks on household spending as observed in the main results, albeit with larger magnitudes. This finding suggests that the refined sample, which better captures areas with stronger attachment to local teams, amplifies the effects of emotional shocks on spending behavior.

4.4 Falsification Tests

We further conducted a falsification test to investigate whether households' shopping decisions in a given week are influenced by the outcomes of NFL games in the following month.³³ The results, presented in [Table A13](#), replicate the benchmark regression from [Table 4](#) (columns 1–3). All point estimates are small in magnitude and not statistically significant at the 5% level. These findings confirm that, as expected, NFL game

³¹Data source: Twitter, [NFL fan map: Where are your team's followers?](#) Accessed on December 1, 2024.

³²Data source: Vividseats, [NFL Fan Map: How America Roots for the NFL](#). Accessed on December 1, 2024.

³³The rationale for using game results from the following month is to rule out potential effects from stockpiling behaviors.

results from the following month have no effect on household spending. Moreover, the lack of statistical significance across all 18 coefficients reported in the table supports the robustness of our main results.³⁴

5 Conclusion

In this paper, we leverage the unexpected outcomes of NFL games as a natural experiment to examine how positive and negative emotional shocks influence household spending in a real-world setting. Using the Las Vegas bookmakers' pregame point spread as a measure of fans' rational expectations about game outcomes, we analyze how unexpected results—such as upset losses, close losses, and upset wins—affect household consumption decisions.

Our findings reveal that emotional shocks from unexpected NFL game outcomes significantly affect household spending, though the effects are short-lived. Negative shocks, such as upset losses, lead to an immediate 2.3% increase in spending on game day (Sunday), which intensifies to 5.2% during Monday-Tuesday before fading over the week. In contrast, positive shocks, such as upset wins, result in a 4.1% reduction in spending during Monday-Tuesday, with no significant effect observed on Sunday. Close losses do not have a measurable impact on spending. These results suggest that negative emotional shocks elicit stronger and more immediate behavioral responses than positive ones. Furthermore, consumers appear to increase spending as a coping mechanism in response to negative emotions, while we find no evidence of spending increases driven by celebratory or contentment-related behavior. Notably, the increase in spending is primarily driven by more frequent shopping trips.

Extending the analysis, we find that food expenditures are more sensitive to emotional shocks than alcohol and non-food items. Spending responses are further ampli-

³⁴We replicated the robustness checks for both all games and home games. The results, presented in Appendix [Table B8](#) to [Table B15](#), are consistent with our main findings.

fied during high-stakes games, and the effects vary across demographic groups. Notably, lower-income households respond more quickly and exhibit consistent reactions to both positive and negative shocks. A series of robustness checks and a falsification test confirm the validity and reliability of these findings.

Extending the analysis, we find that food expenditures are more sensitive to emotional shocks compared to alcohol and non-food items. Spending responses are further amplified during high-stakes games, and the effects vary across demographic groups. Notably, lower-income households respond more quickly and exhibit consistent reactions to both positive and negative shocks. A series of robustness checks and a falsification test confirm the validity and reliability of these findings.

From a broader perspective, this study provides empirical evidence that short-term emotions significantly influence consumption decisions in real-world settings, highlighting their role in shaping economic behavior. Specifically, our work contributes to the limited empirical literature on the causal relationship between emotions and spending, particularly across diverse product categories.

Our findings have important implications for both policymakers and businesses. For policymakers, the results suggest that emotional shocks from unexpected sporting outcomes can trigger significant short-term changes in household consumption patterns, offering insights into household responses to other unforeseen events. For businesses, especially in the retail sector, the findings reveal systematic spending patterns following unexpected game outcomes, particularly in food and non-food categories during the days immediately after games. These patterns suggest opportunities for businesses to better understand and respond to emotion-driven changes in consumer behavior. However, the documented relationship between emotional shocks and spending behavior also raises concerns about consumer vulnerability during periods of heightened emotional states.

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Tables

Table 1: Summary Statistics of Sunday Regular Season NFL Games, 2004-2019

Home & Division Games	Number of Games	Fraction of Category or Subcategory (%)
<i>Panel A. Game Outcomes</i>		
Loss	450	45.59
Win	537	54.41
<i>Panel B. Predicted and Actual Outcomes</i>		
Predicted Win: Point Spread ≤ -4	390	39.51
Actual Win	291	74.62
Actual Loss (Upset Loss)	99	25.38
Predicted Close: $-4 < \text{Point Spread} < 4$	444	44.98
Actual Win	216	48.65
Actual Loss (Close Loss)	228	51.35
Predicted Loss: Point Spread ≥ 4	153	15.50
Actual Win (Upset Win)	30	19.61
Actual Loss	123	80.39
<i>Panel C. Game Time</i>		
Games Starting Before 1 PM	637	64.54
Games Starting Before 7 PM	902	91.39
<i>Panel D. Salience Games</i>		
Sacks ≥ 5 , Turnovers ≥ 5 , or Penalties > 100 yds	538	54.51

Notes: (1) This table presents summary statistics for NFL Sunday home division football games during the first 16 weeks of each regular season from 2004 to 2019. (2) We exclude the New York Jets, New York Giants, Los Angeles Chargers, and Los Angeles Rams due to their shared home stadium locations. (3) Between 2004 and 2019, 5 regular-season Home & Division Games NFL games played on Sundays ended in a tie. These ties were counted as a half-win and half-loss in league standings. We exclude these games in our analysis.

Table 2: Summary Statistics of Household Shopping Behaviors

	Game Week		Bye Week	
	Mean	Std. Dev.	Mean	Std. Dev.
Total Expenditure				
Sunday	17.00	47.81	17.40	47.95
Monday to Tuesday	19.91	43.32	19.90	43.07
Monday to Saturday	67.16	84.90	67.61	84.84
Food Expenditure				
Sunday	11.17	32.67	11.53	33.02
Monday to Tuesday	12.46	28.59	12.63	28.82
Monday to Saturday	42.08	54.07	42.89	54.75
Alcohol Expenditure				
Sunday	0.47	5.15	0.44	5.03
Monday to Tuesday	0.66	5.87	0.61	5.41
Monday to Saturday	2.50	11.92	2.32	11.55
Non-Food Expenditure				
Sunday	5.36	20.92	5.43	20.70
Monday to Tuesday	6.78	22.10	6.67	21.52
Monday to Saturday	22.58	43.64	22.41	42.70
Shopping Trips ^a				
Sunday	0.28	0.45	0.29	0.45
Monday to Tuesday	0.47	0.64	0.47	0.64
Monday to Saturday	1.46	1.33	1.47	1.33
Sample Size	787,724		388,416	

Notes: This table presents the means and standard deviations of household expenditures and shopping trips following Sunday home division games or bye weeks during regular seasons. We deflate expenditures to 2012 dollars using the consumer price index for urban consumers for all items. ^aShopping trips are defined by the frequency of visits made to purchase groceries. Each visit is counted as one trip, regardless of whether it involves stops at multiple stores.

Table 3: Summary Statistics of Household Characteristics

Variable	Description	Mean	Std. Dev
Age	Household head's age	56.21	13.11
Education years	Household head's education years	14.46	1.97
Children	Indicates whether the household has children	0.23	0.42
Household size	The number of household members	2.35	1.29
Married ^a	Indicates whether the household status is married	0.61	0.49
White	Indicates whether the race of the household is white	0.82	0.38
Income ^b	The real income of the household (\$1,000)	58.60	34.01
Internet access	Indicates whether the household's internet is active	0.86	0.35
Number of Households	The unique agents in our sample	83,332	

Notes: The summary statistics are unweighted for broader representativeness, as the final sample cannot be extrapolated to national, regional, or scantrack market levels where projection factors were applied in the HomeScan data. In this table, we provide descriptions and summary statistics for household socio-demographic characteristics. For households with two heads, we report the mean age and education observed for both heads. We deflate incomes to 2012 dollars using the consumer price index for urban consumers for all items. ^aFor single household heads, 67.56% are female and 32.44% are male. ^bWe use income data from two years before the panel year. Using current year income data would significantly reduce the number of observations due to sample attrition. Nonetheless, our results remain consistent when we use the current year income data.

Table 4: The Effects of Emotional Shocks on Household Total Spending and Shopping Trips

<i>Dependent Var:</i>	Ln (Total Spending)			Shopping Trips		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Upset Loss	0.023*** (0.008)	0.052*** (0.008)	0.007 (0.008)	0.013** (0.006)	0.030*** (0.005)	0.000 (0.003)
Close Loss	0.005 (0.007)	0.008 (0.007)	0.005 (0.008)	0.008 (0.005)	0.003 (0.004)	-0.000 (0.003)
Upset Win	-0.013 (0.017)	-0.041** (0.017)	-0.011 (0.019)	-0.011 (0.014)	-0.036*** (0.012)	-0.005 (0.007)
Predicted Win	-0.010* (0.005)	-0.021*** (0.006)	0.012** (0.006)	-0.009** (0.004)	-0.010*** (0.004)	0.002 (0.002)
Predicted Close	0.005 (0.006)	-0.002 (0.006)	0.003 (0.007)	0.005 (0.005)	0.003 (0.004)	0.002 (0.002)
Predicted Loss	0.026*** (0.009)	0.024*** (0.009)	0.025** (0.010)	0.019*** (0.007)	0.014** (0.006)	0.009** (0.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.129	0.152	0.206	-	-	-
Observations	1,176,140	1,176,140	1,176,140	1,080,076	1,102,474	1,160,389

Notes: The sample is restricted to expenditures/shopping trips following Sunday home division games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education years, household size, and real income. In addition, controls include an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends) and the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) in columns (1)-(3) or shopping trips on the last Saturday in columns (4)-(6). *Bye Weeks* is the reference category. Given the incident-based nature of trip data, we instead use the Poisson model (*PPMLHDFE* command in STATA@18) for the number of trips reported by households. Some observations are excluded in columns (4)-(6) because they are either singletons or separated due to fixed effects. *** p<0.01, ** p<0.05, * p<0.1.

Table 5: Shocks from Emotionally Salient Games

<i>Dependent Var:</i> <i>Ln(total spending on...)</i>	More Salient Games			Less Salient Games		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Upset Loss	0.020* (0.011)	0.092*** (0.011)	0.029** (0.012)	0.023** (0.012)	0.028** (0.012)	-0.027** (0.013)
Close Loss	0.005 (0.009)	0.024** (0.010)	0.017* (0.010)	0.001 (0.010)	-0.011 (0.011)	-0.011 (0.012)
Upset Win	-0.073*** (0.025)	-0.115*** (0.06)	-0.016 (0.028)	0.037 (0.024)	-0.013 (0.025)	-0.000 (0.028)
Predicted Win	-0.005 (0.007)	-0.047*** (0.008)	-0.003 (0.008)	-0.009 (0.007)	-0.005 (0.008)	0.029*** (0.008)
Predicted Close	0.007 (0.008)	-0.028*** (0.008)	-0.013 (0.009)	0.013 (0.008)	0.020** (0.009)	0.018** (0.009)
Predicted Loss	0.041*** (0.012)	0.078*** (0.013)	0.059*** (0.014)	0.024** (0.012)	-0.026** (0.013)	-0.006 (0.014)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.130	0.153	0.207	0.129	0.151	0.202
Observations	801,172	801,172	801,172	743,848	743,848	743,848

Notes: This table examines the impact of three potentially frustrating events: five or more sacks, five or more turnovers, or 100 or more penalty yards. The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the reference category. *** p<0.01, ** p<0.05, * p<0.1.

Table 6: Emotional Shocks and Household Total Spending by Income Level, Age, and Race

Dependent Var:	Income Level			Age			Race		
<i>Ln (total spending ...)</i>	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
<i>Panel A</i>	Below Average			<65			Non-white		
(a) Upset Loss	0.027*** (0.009)	0.037*** (0.010)	0.002 (0.010)	0.026*** (0.009)	0.038*** (0.009)	0.006 (0.010)	0.021** (0.008)	0.056*** (0.009)	0.008 (0.009)
(b) Close Loss	-0.008 (0.009)	0.022** (0.009)	0.009 (0.010)	0.004 (0.008)	0.005 (0.008)	0.001 (0.009)	0.005 (0.007)	0.012 (0.008)	0.012 (0.008)
(c) Upset Win	-0.011 (0.020)	-0.048** (0.022)	-0.019 (0.023)	-0.000 (0.020)	-0.036* (0.020)	0.002 (0.023)	-0.022 (0.019)	-0.057*** (0.019)	-0.022 (0.021)
Predicted Win	-0.019*** (0.007)	-0.010 (0.007)	0.021*** (0.008)	-0.014** (0.007)	-0.014** (0.007)	0.007 (0.007)	-0.010* (0.006)	-0.022*** (0.006)	0.009 (0.007)
Predicted Close	0.016** (0.007)	-0.008 (0.008)	0.006 (0.008)	0.005 (0.007)	0.002 (0.007)	0.002 (0.008)	0.003 (0.007)	-0.009 (0.007)	-0.007 (0.007)
Predicted Loss	0.025** (0.011)	0.022* (0.011)	0.034*** (0.012)	0.021** (0.010)	0.027*** (0.010)	0.020* (0.012)	0.027*** (0.010)	0.027*** (0.010)	0.027** (0.011)
Observations	679,514	679,514	679,514	864,445	864,445	864,445	967,026	967,026	967,026
<i>Panel B</i>	Above Average			65+			White		
(a') Upset Loss	0.014 (0.013)	0.074*** (0.013)	0.010 (0.014)	0.009 (0.013)	0.090*** (0.015)	0.007 (0.015)	0.033* (0.018)	0.033* (0.018)	0.000 (0.020)
(b') Close Loss	0.020* (0.011)	-0.013 (0.011)	-0.004 (0.012)	0.010 (0.012)	0.007 (0.014)	0.016 (0.014)	0.007 (0.015)	-0.011 (0.016)	-0.018 (0.017)
(c') Upset Win	-0.019 (0.029)	-0.024 (0.029)	0.005 (0.032)	-0.041 (0.029)	-0.041 (0.034)	-0.037 (0.035)	0.017 (0.036)	0.011 (0.038)	0.019 (0.043)
Predicted Win	0.003 (0.009)	-0.033*** (0.009)	0.003 (0.010)	0.000 (0.009)	-0.038*** (0.011)	0.026** (0.011)	-0.014 (0.013)	-0.019 (0.013)	0.024 (0.015)
Predicted Close	-0.009 (0.010)	0.008 (0.010)	0.003 (0.010)	0.002 (0.010)	-0.007 (0.012)	0.007 (0.012)	0.010 (0.014)	0.029** (0.014)	0.044*** (0.015)
Predicted Loss	0.029** (0.014)	0.026* (0.014)	0.010 (0.016)	0.035** (0.015)	0.018 (0.017)	0.043** (0.018)	0.018 (0.020)	0.012 (0.020)	0.019 (0.023)
Observation	496,444	496,444	496,444	311,660	311,660	311,660	209,102	209,102	209,102
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>p</i> -value for row (a) = row (a')	0.504	0.008	0.261	0.181	0.038	0.956	0.330	0.864	0.761
<i>p</i> -value for row (b) = row (b')	0.078	0.029	0.485	0.665	0.408	0.344	0.562	0.372	0.224
<i>p</i> -value for row (c) = row (c')	0.345	0.766	0.529	0.526	0.912	0.348	0.219	0.006	0.334

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: Emotional Shocks and Household Shopping on Different Types of Food

<i>Dependent Var:</i> <i>Ln (total spending on...)</i>	Perishable Food			Non-perishable Food		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Upset Loss	0.010** (0.005)	0.026*** (0.005)	0.003 (0.006)	0.018*** (0.006)	0.040*** (0.007)	0.010 (0.008)
Close Loss	0.004 (0.004)	0.009* (0.005)	-0.000 (0.006)	0.002 (0.006)	0.010* (0.006)	0.001 (0.007)
Upset Win	-0.021** (0.011)	-0.022** (0.011)	-0.005 (0.014)	-0.024* (0.014)	-0.038*** (0.014)	-0.010 (0.017)
Predicted Win	-0.002 (0.004)	-0.015*** (0.004)	0.010** (0.005)	-0.008* (0.004)	-0.017*** (0.005)	0.008 (0.005)
Predicted Close	0.004 (0.004)	-0.006 (0.004)	0.002 (0.005)	0.003 (0.005)	-0.009* (0.005)	-0.000 (0.006)
Predicted Loss	0.016*** (0.006)	0.009 (0.006)	0.010 (0.007)	0.021*** (0.007)	0.021*** (0.008)	0.024*** (0.009)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.126	0.129	0.206	0.125	0.132	0.192
Observations	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. Perishable food includes dairy, deli, fresh produce, and packaged meat departments. Non-perishable food includes dry grocery and frozen foods departments. These categorizations are based on product hierarchy structured by the NielsenIQ Company. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household marital status, childbearing, race, and internet access, as well as household age, education, household size, and real income. Additional controls include the logarithm of spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the reference category. *** p<0.01, ** p<0.05, * p<0.1.

Figures

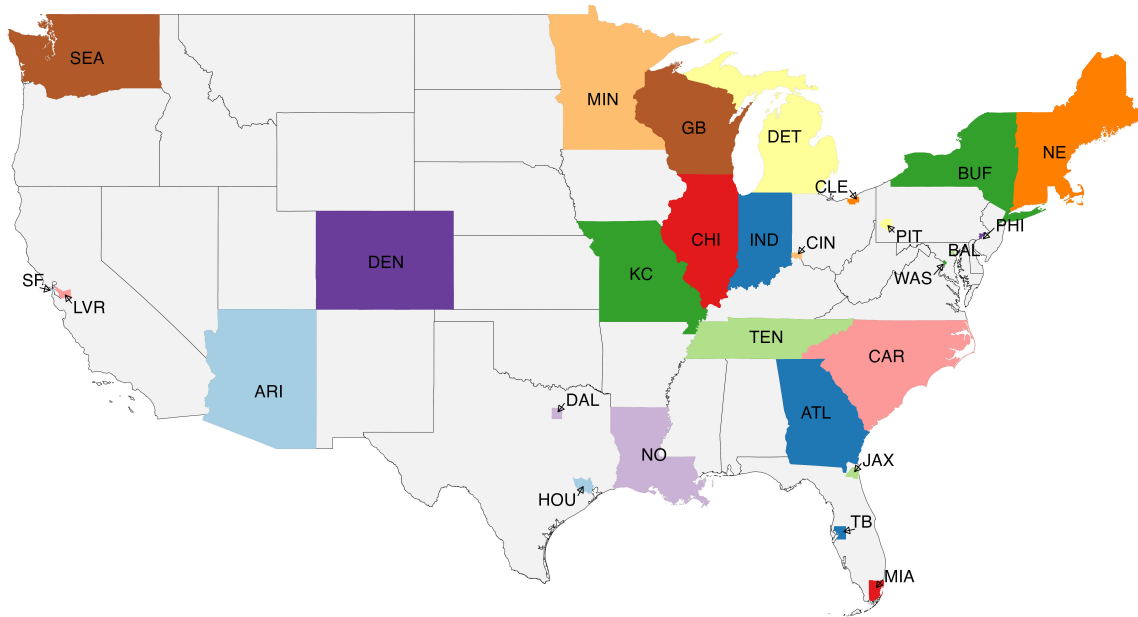


Figure 1: NFL Teams and Fan Map

Notes: This map illustrates the NFL teams included in the study sample and the geographic distribution of local franchises. For states with a single NFL team, all panelists within the state are associated with that team. In states with multiple NFL teams, panelists are assigned to the team corresponding to their city of residence.

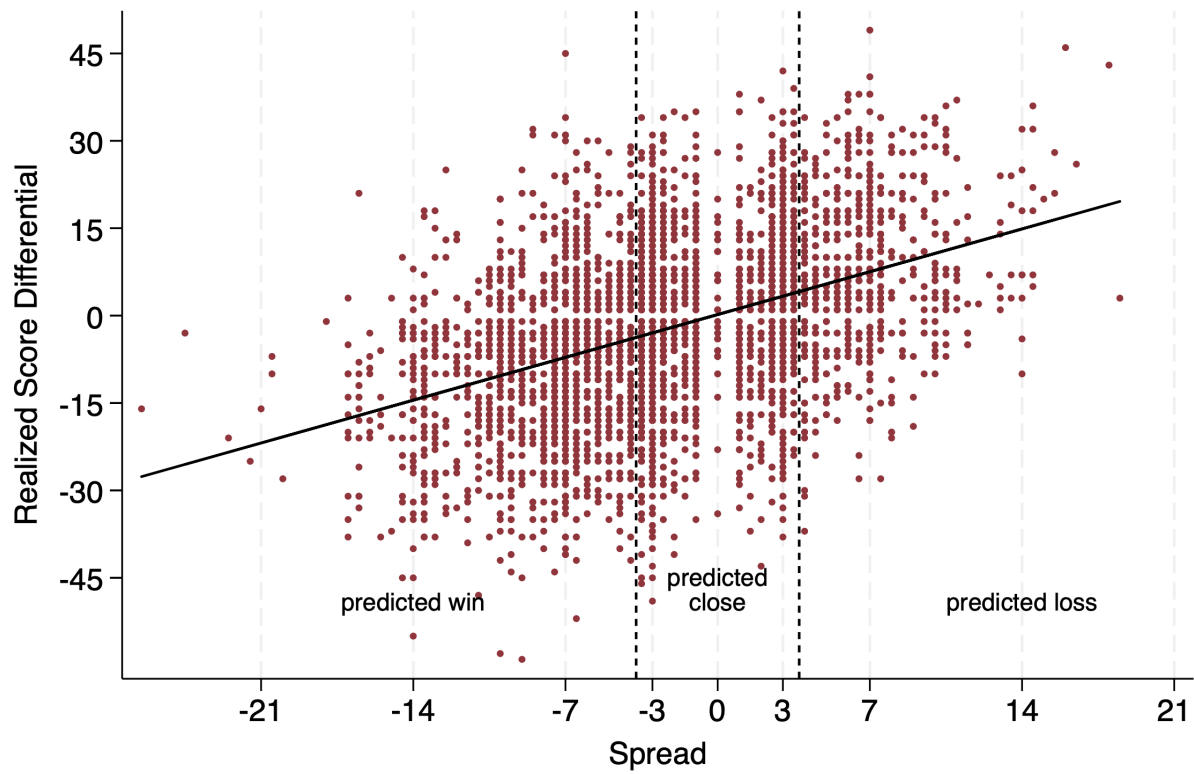


Figure 2: Final Score Differential versus the Pregame Point Spread

Notes: This figure illustrates the relationship between the actual and predicted point spreads for each NFL game during the 2004-2019 seasons. The regression line plotted has a slope of 1.05 (standard error = 0.03).

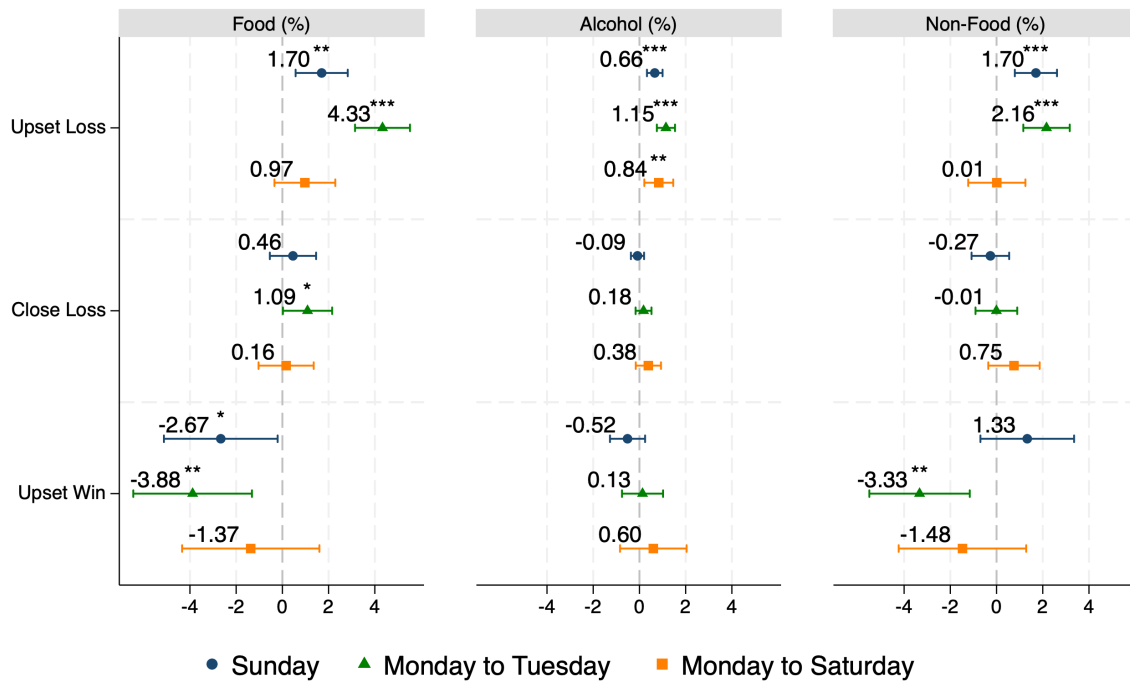


Figure 3: Emotional Shocks and Household Spendings on Different Categories

Notes: This figure illustrates the estimated coefficients for the emotional shocks on household total spending. The bars represent 95 percent confidence intervals, and standard errors are clustered at the household level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. [Table A5](#) provide further information.

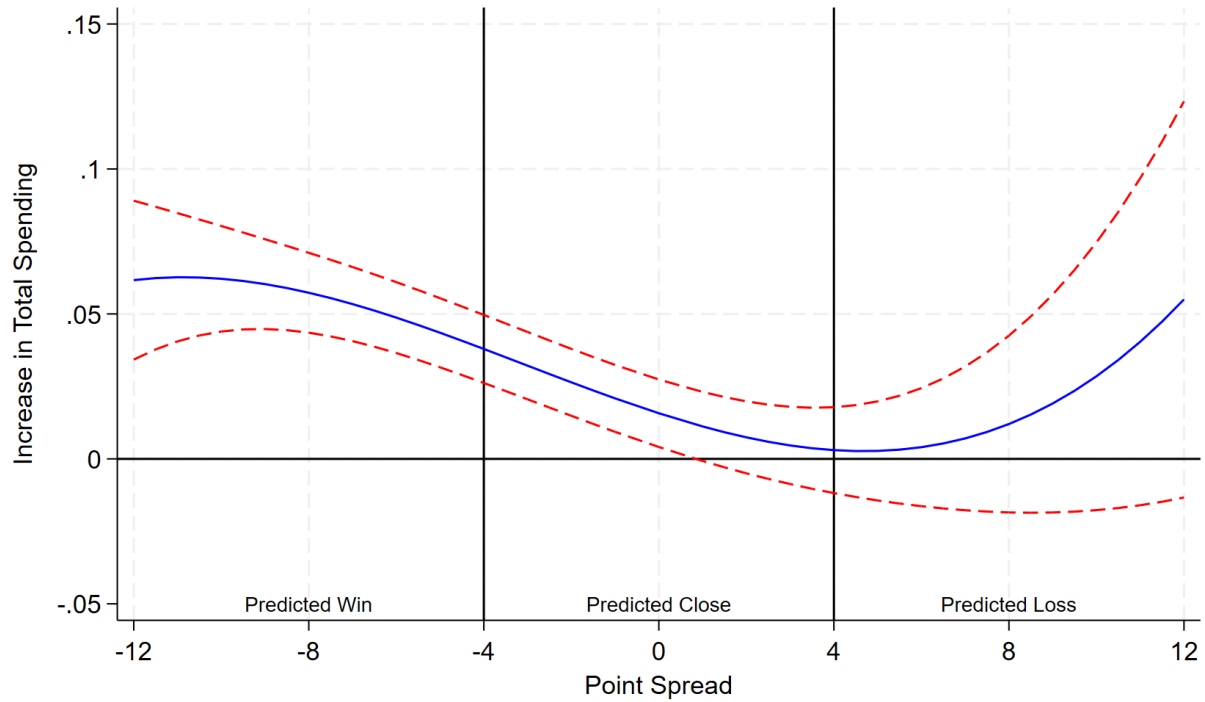


Figure 4: Increase in Total Spending for a Loss versus a Win as a Function of the Pregame Point Spread

Notes: The estimates are derived from a specification with a cubic-order polynomial in the point spread and an indicator variable for team loss. The dependent variable is the household total spending during Monday and Tuesday following a Sunday home division game, or bye week during the regular NFL football season from 2004 to 2019. The dashed lines represent pointwise 95 percent confidence intervals.

Appendix A Appendix for Main Text

Table A1: NFL Football Teams

Conference	Division	Team	Team ID
AFC	EAST	Buffalo Bills	BUF
		Miami Dolphins	MIA
		New England Patriots	NE
		New York Jets	NYJ
	NORTH	Baltimore Ravens	BAL
		Cincinnati Bengals	CIN
		Cleveland Browns	CLE
		Pittsburgh Steelers	PIT
	SOUTH	Houston Texans	HOU
		Indianapolis Colts	IND
		Jacksonville Jaguars	JAX
		Tennessee Titans	TEN
	WEST	Denver Broncos	DEN
		Kansas City Chiefs	KC
		Los Angeles Chargers	LAC
		Oakland Raiders	LVR
NFC	EAST	Dallas Cowboys	DAL
		New York Giants	NYG
		Philadelphia Eagles	PHI
		Washington Redskins	WAS
	NORTH	Chicago Bears	CHI
		Detroit Lions	DET
		Green Bay Packers	GB
		Minnesota Vikings	MIN
	SOUTH	Atlanta Falcons	ATL
		Carolina Panthers	CAR
		New Orleans Saints	NO
		Tampa Bay Buccaneers	TB
	WEST	Arizona Cardinals	ARI
		Los Angeles Rams	LAR
		San Francisco 49ers	SF
		Seattle Seahawks	SEA

Notes: The United States has two major professional football conferences: the American Football Conference (AFC) and the National Football Conference (NFC). Each conference is divided into four divisions, with each division comprising four teams.

Table A2: NFL Football Teams Matched to Households in Sample (2004 - 2019)

Team	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Arizona Cardinals	1,430	1,371	1,608	1,768	1,671	1,615	1,694	1,642	1,605	1,618	1,546	1,542	1,570	1,531	1,512	1,486
Atlanta Falcons	1,349	1,342	1,764	1,950	1,925	1,921	2,035	1,981	1,941	2,049	2,089	2,127	2,116	2,064	2,129	2,203
Baltimore Ravens	76	77	107	109	109	115	116	111	116	119	115	114	114	114	111	117
Buffalo Bills	326	305	341	356	347	333	318	287	266	252	243	231	225	207	207	203
Carolina Panthers	2,020	1,968	3,185	3,521	3,483	3,552	3,754	3,548	3,492	3,560	3,530	3,535	3,531	3,619	3,686	3,594
Chicago Bears	2,040	2,005	2,819	3,093	3,152	3,389	3,416	3,174	3,154	3,172	3,134	3,135	3,224	3,069	2,976	3,007
Cincinnati Bengals	69	65	200	222	208	212	227	206	204	207	215	218	215	210	221	201
Cleveland Browns	77	71	147	158	174	176	171	163	170	176	179	175	164	165	169	172
Dallas Cowboys	151	151	165	172	166	176	188	169	169	166	175	184	180	173	167	168
Denver Broncos	1,310	1,275	1,557	1,634	1,495	1,441	1,500	1,390	1,348	1,361	1,350	1,348	1,405	1,377	1,341	1,302
Detroit Lions	1,448	1,474	2,436	2,726	2,623	2,722	2,761	2,609	2,611	2,610	2,580	2,651	2,698	2,643	2,600	2,570
Green Bay Packers	621	631	1,547	1,751	1,686	1,643	1,672	1,608	1,612	1,679	1,704	1,665	1,667	1,649	1,613	1,593
Houston Texans	542	510	570	578	529	525	539	519	513	535	547	550	554	528	523	502
Indianapolis Colts	591	588	1,604	1,805	1,723	1,738	1,721	1,667	1,640	1,670	1,685	1,696	1,715	1,705	1,692	1,648
Jacksonville Jaguars	60	57	163	177	165	160	161	150	152	171	174	191	182	175	178	169
Kansas City Chiefs	46	49	100	124	132	132	157	161	165	171	165	169	165	170	168	185
Miami Dolphins	241	250	252	241	242	252	243	242	248	249	236	241	239	223	222	225
Minnesota Vikings	1,427	1,409	1,749	1,857	1,798	1,708	1,769	1,654	1,589	1,611	1,544	1,557	1,552	1,480	1,458	1,440
New England Patriots	2,054	1,938	2,901	3,176	3,160	3,283	3,340	3,202	3,257	3,340	3,355	3,408	3,476	3,376	3,310	3,298
New Orleans Saints	696	588	851	877	826	849	841	802	815	849	860	878	918	924	949	933
Oakland Raiders	78	75	78	76	71	76	80	71	70	71	64	67	62	57	54	54
Philadelphia Eagles	202	186	213	233	223	239	248	245	248	258	248	269	272	250	251	268
Pittsburgh Steelers	80	77	216	242	226	222	225	222	220	220	221	207	217	203	206	207
San Diego Chargers	77	67	236	254	229	237	255	232	224	235	228	238	206	-	-	-
San Francisco 49ers	161	149	149	145	134	134	137	137	130	133	138	136	140	134	128	126
Seattle Seahawks	1,520	1,467	1,822	1,913	1,810	1,747	1,762	1,720	1,693	1,684	1,698	1,693	1,675	1,590	1,532	1,462
St. Louis Rams	524	498	513	515	508	463	465	419	399	376	340	302	-	-	-	-
Tampa Bay Buccaneers	134	133	154	175	180	165	169	163	162	153	148	169	155	148	153	158
Tennessee Titans	1,012	1,068	1,584	1,683	1,564	1,573	1,618	1,541	1,531	1,558	1,586	1,611	1,623	1,629	1,606	1,574
Washington Redskins	86	81	93	91	86	90	86	80	79	87	91	89	91	83	82	89
Total	20,448	19,925	29,124	31,622	30,645	30,888	31,668	30,115	29,823	30,340	30,188	30,396	30,351	29,496	29,244	28,954

Table A3: The Effects of Emotional Shocks on Household Total Spending and Shopping Trips (Full Table)

<i>Dependent Var:</i>	Ln (Total Spending)			Shopping Trips		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
(a)Upset Loss	0.023*** (0.008)	0.052*** (0.008)	0.007 (0.008)	0.013** (0.006)	0.030*** (0.005)	0.000 (0.003)
Close Loss	0.005 (0.007)	0.008 (0.007)	0.005 (0.008)	0.008 (0.005)	0.003 (0.004)	-0.000 (0.003)
(b)Upset Win	-0.013 (0.017)	-0.041** (0.017)	-0.011 (0.019)	-0.011 (0.014)	-0.036*** (0.012)	-0.005 (0.007)
Predicted Win	-0.010* (0.005)	-0.021*** (0.006)	0.012** (0.006)	-0.009** (0.004)	-0.010*** (0.004)	0.002 (0.002)
Predicted Close	0.005 (0.006)	-0.002 (0.006)	0.003 (0.007)	0.005 (0.005)	0.003 (0.004)	0.002 (0.002)
Predicted Loss	0.026*** (0.009)	0.024*** (0.009)	0.025** (0.010)	0.019*** (0.007)	0.014** (0.006)	0.009** (0.004)
Ln(Total Spending on Last Sat)/ Trips on Last Sat	-0.078*** (0.001)	-0.032*** (0.001)	0.018*** (0.001)	-0.169*** (0.004)	-0.028*** (0.003)	0.042*** (0.002)
Age	-0.000 (0.001)	0.001 (0.001)	0.003** (0.002)	-0.001 (0.001)	0.000 (0.001)	0.001* (0.001)
Education years	0.000 (0.004)	0.005 (0.004)	0.005 (0.005)	-0.001 (0.003)	0.002 (0.003)	0.000 (0.002)
Children	0.035** (0.014)	0.093*** (0.015)	0.084*** (0.017)	0.018* (0.010)	0.061*** (0.009)	0.042*** (0.006)
Household Size	0.020*** (0.005)	0.020*** (0.006)	0.025*** (0.006)	0.011*** (0.004)	0.009** (0.003)	0.007*** (0.002)
Married	0.047*** (0.015)	0.135*** (0.016)	0.185*** (0.018)	0.035*** (0.012)	0.080*** (0.010)	0.080*** (0.007)
White	0.043 (0.028)	-0.028 (0.029)	-0.031 (0.032)	0.029 (0.021)	-0.016 (0.019)	-0.010 (0.013)
Ln(income)	0.038*** (0.007)	-0.028*** (0.008)	-0.022** (0.009)	0.022*** (0.006)	-0.024*** (0.005)	-0.014*** (0.004)
Internet access	-0.011 (0.012)	-0.013 (0.013)	-0.025* (0.014)	-0.015 (0.009)	-0.008 (0.008)	-0.007 (0.006)
Holiday week	-0.019*** (0.006)	-0.044*** (0.007)	0.013* (0.007)	-0.012** (0.005)	-0.027*** (0.004)	0.007*** (0.003)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.129	0.152	0.206	-	-	-
Observations	1,176,140	1,176,140	1,176,140	1,080,076	1,102,474	1,160,389

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. *Bye Weeks* is the reference category. Some observations are excluded in columns (4)-(6) because they are either singletons or separated due to fixed effects. *** p<0.01, ** p<0.05, * p<0.1.

Table A4: Emotional Shocks and Household Spendings on Different Categories

<i>Dependent Var:</i>	Food Spending			Alcohol Spending			Non-Food Spending		
<i>Ln(spending on...)</i>	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.017** (0.007)	0.043*** (0.007)	0.010 (0.008)	0.007*** (0.002)	0.011*** (0.002)	0.008** (0.004)	0.017*** (0.006)	0.022*** (0.006)	0.000 (0.008)
Close Loss	0.005 (0.006)	0.011* (0.006)	0.002 (0.007)	-0.001 (0.002)	0.002 (0.002)	0.004 (0.003)	-0.003 (0.005)	-0.000 (0.005)	0.008 (0.007)
Upset Win	-0.027* (0.015)	-0.039** (0.016)	-0.014 (0.018)	-0.005 (0.005)	0.001 (0.005)	0.006 (0.009)	0.013 (0.012)	-0.033** (0.013)	-0.015 (0.017)
Predicted Win	-0.007 (0.005)	-0.020*** (0.005)	0.011* (0.006)	-0.001 (0.001)	-0.007*** (0.002)	-0.009*** (0.003)	-0.009** (0.004)	-0.011** (0.004)	0.007 (0.005)
Predicted Close	0.005 (0.005)	-0.008 (0.006)	0.003 (0.006)	0.001 (0.002)	-0.003* (0.002)	-0.001 (0.003)	0.002 (0.004)	0.000 (0.005)	-0.005 (0.006)
Predicted Loss	0.023*** (0.008)	0.021*** (0.008)	0.026*** (0.009)	0.005** (0.002)	0.003 (0.003)	0.012*** (0.004)	0.008 (0.006)	0.015** (0.007)	0.015* (0.009)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.129	0.143	0.200	0.088	0.129	0.277	0.096	0.114	0.195
Observations	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. Food includes dry grocery, frozen foods, dairy, deli, fresh produce, and packaged meat departments. Non-food includes health & beauty care, non-food grocery, and general merchandise departments. These categorizations are from the consumer panel data collected by the NielsenIQ Company. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Timing of Shocks and Spending on Sunday

<i>Dependent Var:</i>	(1)	(2)
<i>Ln (total spending on Sunday)</i>	Games before 1pm	Games before 7pm
Upset Loss	0.025*** (0.009)	0.028*** (0.008)
Close Loss	0.009 (0.008)	0.003 (0.007)
Upset Win	-0.012 (0.019)	-0.016 (0.017)
Predicted Win	-0.011* (0.006)	-0.010* (0.006)
Predicted Close	0.004 (0.007)	0.005 (0.006)
Predicted Loss	0.028*** (0.010)	0.022** (0.009)
Controls	Yes	Yes
Household FE	Yes	Yes
Season & Week of season FE	Yes	Yes
Adj. R^2	0.130	0.129
Observations	931,140	1,126,228

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household marital status, childbearing, race, and internet access, as well as household age, education, household size, and real income. Additional controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the reference category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Emotional Shocks and Household Total Spending - Different Cutoffs

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.023*** (0.007)	0.049*** (0.007)	0.013* (0.007)
Close Loss	0.010 (0.011)	0.010 (0.011)	-0.010 (0.012)
Upset Win	0.011 (0.010)	-0.017 (0.011)	-0.012 (0.012)
Predicted Win	-0.008 (0.005)	-0.016*** (0.005)	0.009* (0.006)
Predicted Close	-0.004 (0.008)	-0.008 (0.008)	0.006 (0.009)
Predicted Loss	0.020*** (0.007)	0.006 (0.007)	0.016** (0.008)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj.R ²	0.129	0.152	0.206
Observations	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -3 or less, *Predicted Close* indicates a point spread between -3 and 3 (exclusive), and *Predicted Loss* indicates a spread of +3 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table A7: Emotional Shocks and Household Total Spending - Different Fixed Effects

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.025*** (0.009)	0.022** (0.010)	0.013 (0.010)
Close Loss	0.009 (0.008)	0.013 (0.008)	0.013 (0.009)
Upset Win	-0.033 (0.020)	-0.011 (0.021)	0.004 (0.023)
Predicted Win	-0.005 (0.006)	-0.001 (0.007)	0.015** (0.007)
Predicted Close	-0.000 (0.007)	-0.001 (0.007)	-0.004 (0.008)
Predicted Loss	0.025** (0.011)	0.021* (0.011)	0.022* (0.012)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season - week FE	Yes	Yes	Yes
Adj.R ²	0.130	0.154	0.206
Observations	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category.

*** p<0.01, ** p<0.05, * p<0.1.

Table A8: Emotional Shocks and Household Total Spending - Control for Last Week Spending

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.022*** (0.008)	0.051*** (0.008)	0.007 (0.008)
Close Loss	0.004 (0.007)	0.008 (0.007)	0.006 (0.008)
Upset Win	-0.017 (0.017)	-0.042** (0.017)	-0.010 (0.019)
Predicted Win	-0.010* (0.005)	-0.020*** (0.006)	0.013** (0.006)
Predicted Close	0.006 (0.006)	-0.002 (0.006)	0.002 (0.007)
Predicted Loss	0.025*** (0.009)	0.023*** (0.009)	0.025** (0.010)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.137	0.152	0.209
Observations	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last week and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Emotional Shocks and Household Total Spending - Clustering at HH and season level

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
upset_loss	0.023*** (0.008)	0.052*** (0.008)	0.007 (0.009)
close_loss	0.005 (0.007)	0.008 (0.007)	0.005 (0.008)
upset_win	-0.013 (0.017)	-0.041** (0.018)	-0.011 (0.019)
Predicted Win	-0.010* (0.006)	-0.021*** (0.006)	0.012** (0.006)
Predicted Close	0.005 (0.006)	-0.002 (0.006)	0.003 (0.007)
Predicted Loss	0.026*** (0.009)	0.024*** (0.009)	0.025** (0.010)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.129	0.152	0.206
Observations	1,176,140	1,176,140	1,176,140

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household and game season level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Emotional Shocks and Household Total Spending - Exclude HHs with over a month of zero spending

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.023*** (0.008)	0.053*** (0.009)	0.009 (0.009)
Close Loss	0.009 (0.007)	0.011 (0.008)	0.011 (0.008)
Upset Win	-0.001 (0.018)	-0.046** (0.019)	-0.012 (0.020)
Predicted Win	-0.011** (0.006)	-0.022*** (0.006)	0.009 (0.006)
Predicted Close	0.002 (0.006)	-0.004 (0.007)	-0.003 (0.007)
Predicted Loss	0.021** (0.009)	0.029*** (0.010)	0.025** (0.010)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.137	0.151	0.186
Observations	1,064,209	1,064,209	1,064,209

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household and game season level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A11: Emotional Shocks and Household Total Spending - Different Methods for Last Week's Shopping Data

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.025*** (0.007)	0.035*** (0.007)	-0.004 (0.008)
Close Loss	0.001 (0.006)	0.000 (0.007)	0.001 (0.007)
Upset Win	-0.010 (0.015)	-0.043*** (0.016)	0.024 (0.017)
Predicted Win	-0.008 (0.005)	-0.021*** (0.005)	0.012** (0.006)
Predicted Close	0.002 (0.006)	0.000 (0.006)	-0.000 (0.006)
Predicted Loss	0.021*** (0.008)	0.028*** (0.008)	-0.001 (0.009)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.138	0.152	0.186
Observations	1,345,320	1,345,320	1,345,320

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: Emotional Shocks and Household Total Spending - Team Fans Defined at the County Level

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Loss	0.032** (0.013)	0.070*** (0.013)	0.012 (0.014)
Close Loss	0.013 (0.011)	0.010 (0.011)	-0.003 (0.012)
Upset Win	0.004 (0.025)	-0.078*** (0.026)	0.016 (0.029)
Predicted Win	-0.016* (0.009)	-0.017* (0.009)	0.004 (0.010)
Predicted Close	-0.001 (0.009)	0.008 (0.009)	0.023** (0.010)
Predicted Loss	0.033** (0.013)	0.027** (0.014)	0.022 (0.015)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.132	0.154	0.208
Observations	474,310	474,310	474,310

Notes: The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for households' marital status, childbearing, race, and internet access, as well as households' age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A13: Emotional Shocks and Household Total Spending - Falsification Test

Dependent Var: Ln (spending on...)	(1) Sunday	(2) MonToTue	(3) MonToSat
Upset Losss-following month	-0.005 (0.009)	0.018* (0.009)	0.015 (0.010)
Close Losss-following month	-0.003 (0.006)	0.001 (0.006)	0.009 (0.007)
Upset Wins-following month	0.015 (0.010)	0.016 (0.010)	-0.013 (0.011)
Predicted Win	-0.011 (0.012)	-0.019 (0.013)	-0.004 (0.014)
Predicted Close	-0.014 (0.012)	-0.007 (0.012)	-0.011 (0.013)
Predicted Loss	-0.016 (0.012)	-0.009 (0.013)	0.002 (0.014)
Controls	Yes	Yes	Yes
Household FE	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes
Adj. R^2	0.203	0.224	0.206
Observations	1,007,709	1,007,709	1,007,709

Notes: In this table, we investigated whether households' shopping decisions in a given week are impacted by the game results of the following month as a falsification test. The sample is restricted to expenditures following Sunday home division games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for households' marital status, childbearing, race, and internet access, as well as households' age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix B Results for ALL Games and Home Games

Table B1: Summary Statistics of NFL Games, 2004-2019

	Number of Games	Fraction on Category or Subcategory (%)
<i>Panel A. Sunday Regular Season NFL Games</i>		
All Games	6,072	
Games Starting Before 1 PM	3,877	63.85
Games Starting Before 7 PM	5,610	92.39
Home Games	3,027	
Games Starting Before 1 PM	1,918	63.36
Games Starting Before 7 PM	2,795	92.34
<i>Panel B. Sunday Regular Season NFL Games - All Games</i>		
Outcome		
Loss	3,035	49.98
Win	3,037	50.02
Predicted and Actual Outcomes		
Predicted Win: Point Spread ≤ -4	1,650	27.17
Actual Win	1,222	74.06
Actual Loss (Upset Loss)	428	25.94
Predicted Close: $-4 < \text{Point Spread} < 4$	2,773	45.67
Actual Win	1,386	49.98
Actual Loss (Close Loss)	1,387	50.02
Predicted Loss: Point Spread ≥ 4	1,649	27.16
Actual Win (Upset Win)	429	26.02
Actual Loss	1,220	73.98
Salience Games		
(b) Sacks ≥ 5 , Turnovers ≥ 5 , or Penalties > 100 yds	3,212	52.90
<i>Panel C. Sunday Regular Season NFL Games - Home Games</i>		
Outcome		
Loss	1,302	43.01
Win	1,725	56.99
Predicted and Actual Outcomes		
Predicted Win: Point Spread ≤ -4	1,199	39.61
Actual Win	896	74.73
Actual Loss (Upset Loss)	303	25.27
Predicted Close: $-4 < \text{Point Spread} < 4$	1,386	45.79
Actual Win	709	51.15
Actual Loss (Close Loss)	677	48.85
Predicted Loss: Point Spread ≥ 4	442	14.60
Actual Win (Upset Win)	120	27.15
Actual Loss	322	72.85
Salience Games		
(b) Sacks ≥ 5 , Turnovers ≥ 5 , or Penalties > 100 yds	1,602	52.92

Notes: (1) This table presents summary statistics for NFL football games during the first 16 weeks of each season from 2004 to 2019. For each NFL game where both teams are included in the analysis, the game is recorded twice: once from the perspective of Team A and once from the perspective of Team B. (2) We exclude the New York Jets, New York Giants, Los Angeles Chargers, and Los Angeles Rams due to their shared home stadium locations. (3) Between 2004 and 2019, 18 regular-season NFL games played on Sundays ended in a tie. These ties were counted as a half-win and half-loss in league standings. We exclude these games in our analysis.

Table B2: Summary Statistics of Shopping Behaviors, 2004-2019

	All Game Week		Home Game Week		Bye Week	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Total Expenditure						
Sunday	17.12	48.22	17.09	48.09	17.39	47.94
Monday to Tuesday	20.14	43.59	20.11	43.47	19.90	43.07
Monday to Saturday	67.70	85.67	67.60	85.52	67.59	84.83
Food Expenditure						
Sunday	11.23	32.85	11.23	32.84	11.53	33.02
Monday to Tuesday	12.62	28.81	12.62	28.78	12.62	28.81
Monday to Saturday	42.41	54.48	42.36	54.39	42.88	54.75
Alcohol Expenditure						
Sunday	0.46	5.01	0.46	5.04	0.44	5.03
Monday to Tuesday	0.67	5.85	0.67	5.84	0.61	5.41
Monday to Saturday	2.49	12.05	2.49	11.94	2.31	11.54
Non-Food Expenditure						
Sunday	5.42	21.44	5.40	21.29	5.43	20.69
Monday to Tuesday	6.85	22.25	6.83	22.13	6.67	21.52
Monday to Saturday	22.80	44.04	22.75	44.08	22.40	42.69
Shopping Trips ^a						
Sunday	0.28	0.45	0.28	0.45	0.29	0.45
Monday to Tuesday	0.47	0.64	0.47	0.64	0.46	0.64
Monday to Saturday	1.46	1.34	1.46	1.34	1.46	1.33
Sample Size	4,955,746		2,459,866		388,839	

Notes: The summary statistics are unweighted for broader representativeness, as the final sample cannot be extrapolated to national, regional, or scantrack market levels where projection factors were applied in the HomeScan data. This table primarily presents the means and standard deviations of household expenditures and shopping trips following Sunday all games, home games, or bye weeks during regular seasons. We deflate expenditures to 2012 dollars using the consumer price index for urban consumers for all items. ^aShopping trips are defined by the frequency of visits made to purchase groceries. Each visit is counted as one trip, regardless of whether it involves stops at multiple stores.

Table B3: The Effects of Emotional Shocks on Household Total Spendings

<i>Dependent Var:</i> <i>Ln (total spending on...)</i>	All Games			Home Games		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
(a) Upset Loss	0.005 (0.004)	0.010*** (0.004)	-0.001 (0.004)	0.009** (0.004)	0.016*** (0.005)	0.003 (0.005)
Close Loss	0.004* (0.003)	0.003 (0.003)	0.002 (0.003)	0.011*** (0.004)	0.010*** (0.004)	0.007* (0.004)
(b) Upset Win	-0.002 (0.004)	-0.014*** (0.004)	-0.012*** (0.004)	-0.007 (0.008)	-0.020** (0.008)	-0.025*** (0.009)
Predicted Win	-0.010*** (0.004)	-0.007* (0.004)	0.007* (0.004)	-0.010** (0.004)	-0.010** (0.004)	0.008** (0.004)
Predicted Close	-0.006* (0.003)	0.002 (0.004)	0.002 (0.004)	-0.010*** (0.004)	0.003 (0.004)	0.002 (0.004)
Predicted Loss	-0.003 (0.004)	0.008** (0.004)	0.008* (0.004)	0.007 (0.005)	0.016*** (0.006)	0.015** (0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>p</i> -value for row (a) = -row (b)	0.605	0.490	0.028	0.844	0.549	0.032
Adj. R^2	0.130	0.158	0.207	0.130	0.157	0.206
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's childbearing, marital status, race, and internet access, as well as household's age, education years, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the reference category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B4: Emotional Shocks and Household Shopping Trips

<i>Dependent Var: shopping trips</i>	All Games			Home Games		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
Upset Loss	0.002 (0.003)	0.006** (0.002)	-0.000 (0.001)	0.005 (0.003)	0.008*** (0.003)	-0.001 (0.002)
Close Loss	0.003 (0.002)	0.002 (0.002)	0.001 (0.001)	0.009*** (0.003)	0.005** (0.002)	0.002 (0.002)
Upset Win	-0.002 (0.003)	-0.008*** (0.003)	-0.004*** (0.002)	-0.008 (0.006)	-0.014*** (0.005)	-0.007** (0.003)
Predicted Win	-0.008*** (0.003)	-0.004* (0.002)	0.001 (0.001)	-0.007** (0.003)	-0.005* (0.003)	0.002 (0.002)
Predicted Close	-0.006** (0.003)	0.003 (0.002)	0.001 (0.001)	-0.009*** (0.003)	0.004 (0.003)	0.002 (0.002)
Predicted Loss	-0.003 (0.003)	0.006** (0.002)	0.004** (0.002)	0.003 (0.004)	0.009** (0.004)	0.006*** (0.002)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,268,701	5,273,141	5,338,184	2,759,147	2,774,102	2,839,308

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the first 16 weeks of the regular NFL football season from 2004 to 2019. Given the incident-based nature of trip data, we use the Poisson model (PPMLHDFE command in STATA 18) for the number of trips reported by households. Some observations are excluded because they are either singletons or separated due to fixed effects. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (e.g., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B5: Emotional Shocks and Household Spendings on Different Categories

<i>Dependent Var:</i> <i>Ln(spending on...)</i>	All Games			Home Games		
	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
<i>Panel A. Food Spending</i>						
Upset Loss	0.003 (0.003)	0.007** (0.003)	-0.001 (0.004)	0.007* (0.004)	0.014*** (0.004)	0.004 (0.005)
Close Loss	0.004* (0.002)	0.003 (0.002)	0.002 (0.003)	0.009*** (0.003)	0.011*** (0.004)	0.005 (0.004)
Upset Win	-0.003 (0.004)	-0.014*** (0.004)	-0.016*** (0.004)	-0.009 (0.007)	-0.020*** (0.008)	-0.029*** (0.009)
Predicted Win	-0.008** (0.003)	-0.007* (0.003)	0.007* (0.004)	-0.007** (0.003)	-0.009*** (0.004)	0.008** (0.004)
Predicted Close	-0.006* (0.003)	-0.001 (0.003)	0.003 (0.004)	-0.009** (0.004)	-0.001 (0.004)	0.003 (0.004)
Predicted Loss	-0.003 (0.003)	0.006* (0.004)	0.009** (0.004)	0.006 (0.005)	0.014*** (0.005)	0.018*** (0.006)
<i>Panel B. Alcohol Spending</i>						
Upset Loss	0.001 (0.001)	0.000 (0.001)	-0.001 (0.002)	0.002 (0.001)	0.003** (0.001)	0.001 (0.002)
Close Loss	0.001* (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	-0.001 (0.002)
Upset Win	0.001 (0.001)	-0.002** (0.001)	-0.003 (0.002)	-0.003 (0.002)	-0.003 (0.003)	-0.010** (0.004)
Predicted Win	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.002)	-0.000 (0.001)	-0.003** (0.001)	-0.003* (0.002)
Predicted Close	-0.001 (0.001)	-0.001 (0.001)	-0.000 (0.002)	-0.001 (0.001)	-0.000 (0.001)	0.002 (0.002)
Predicted Loss	-0.001 (0.001)	0.000 (0.001)	-0.001 (0.002)	0.002 (0.001)	0.003 (0.002)	0.005** (0.003)
<i>Panel C. Non-food Spending</i>						
Upset Loss	0.004 (0.003)	0.005* (0.003)	-0.002 (0.004)	0.007** (0.003)	0.005 (0.003)	-0.000 (0.004)
Close Loss	0.002 (0.002)	0.001 (0.002)	0.003 (0.003)	0.007** (0.003)	0.004 (0.003)	0.007* (0.004)
Upset Win	0.001 (0.004)	-0.005* (0.003)	-0.002 (0.004)	0.004 (0.007)	-0.010 (0.007)	-0.017** (0.008)
Predicted Win	-0.006** (0.003)	-0.002 (0.003)	0.001 (0.004)	-0.005 (0.003)	-0.005* (0.003)	0.001 (0.004)
Predicted Close	-0.002 (0.003)	0.001 (0.002)	0.001 (0.003)	-0.002 (0.003)	0.002 (0.003)	0.003 (0.003)
Predicted Loss	0.003 (0.003)	0.004* (0.003)	0.008* (0.004)	0.007 (0.004)	0.009** (0.004)	0.013** (0.005)
Observations	5,344,585	5,344,585	5,344,585	5,344,585	5,344,585	5,344,585

Notes: Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the first 16 weeks of regular NFL football season from 2004 to 2019. Food includes dry grocery, frozen foods, dairy, deli, fresh produce, and packaged meat departments. Non-food includes health & beauty care, non-food grocery, and general merchandise departments. These categorizations are from the consumer panel data collected by the NielsenIQ Company. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B6: Shocks from Emotionally Salient Games

<i>Dependent Var:</i>	All Games			Home Games		
<i>Ln(spending on...)</i>	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
<i>Panel A. More Salient Games</i>						
Upset Loss	0.005 (0.005)	0.017*** (0.005)	-0.001 (0.006)	0.007 (0.006)	0.030*** (0.006)	0.006 (0.007)
Close Loss	0.001 (0.003)	0.007** (0.004)	0.001 (0.004)	0.007 (0.005)	0.005 (0.005)	0.012** (0.006)
Upset Win	0.001 (0.006)	-0.011* (0.006)	0.004 (0.006)	-0.023* (0.012)	-0.039*** (0.013)	-0.012 (0.013)
Predicted Win	-0.012*** (0.004)	-0.010** (0.004)	0.008* (0.005)	-0.010** (0.005)	-0.016*** (0.005)	0.005 (0.005)
Predicted Close	-0.002 (0.004)	-0.005 (0.004)	-0.000 (0.004)	-0.005 (0.005)	-0.005 (0.005)	-0.003 (0.005)
Predicted Loss	0.000 (0.005)	0.007 (0.005)	0.006 (0.005)	0.016** (0.007)	0.028*** (0.008)	0.023*** (0.008)
Observations	2,955,412	2,955,412	2,955,412	1,674,715	1,674,715	1,674,715
<i>Panel B. Less Salient Games</i>						
Upset Loss	0.004 (0.005)	0.004 (0.005)	-0.000 (0.006)	0.010 (0.006)	0.004 (0.007)	-0.001 (0.007)
Close Loss	0.006* (0.004)	-0.002 (0.004)	0.003 (0.004)	0.016*** (0.005)	0.015** (0.006)	-0.001 (0.006)
Upset Win	-0.001 (0.006)	-0.020*** (0.006)	-0.025*** (0.006)	0.013 (0.011)	-0.018 (0.012)	-0.025* (0.013)
Predicted Win	-0.005 (0.005)	-0.005 (0.005)	0.009* (0.005)	-0.008 (0.005)	-0.005 (0.005)	0.015*** (0.006)
Predicted Close	-0.008* (0.005)	0.007 (0.005)	0.009* (0.005)	-0.012** (0.005)	0.006 (0.006)	0.014** (0.006)
Predicted Loss	-0.003 (0.005)	0.010** (0.005)	0.012** (0.005)	0.002 (0.007)	0.006 (0.008)	0.007 (0.008)
Observations	2,768,599	2,768,599	2,768,599	1,549,115	1,549,115	1,549,115
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the impact of three potentially frustrating events: five or more sacks, five or more turnovers, or 100 or more penalty yards. The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B7: Emotional Shocks and Household Shopping on Different Types of Food

<i>Dependent Var: Ln (spending on...)</i>	All Games			Home Games		
	(1) Sunday	(2) MonToTue	(3) MonToSat	(4) Sunday	(5) MonToTue	(6) MonToSat
<i>Panel A. Perishable Food</i>						
Upset Loss	0.002 (0.002)	0.003 (0.002)	-0.000 (0.003)	0.006** (0.003)	0.010*** (0.003)	0.003 (0.004)
Close Loss	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.007*** (0.002)	0.010*** (0.003)	0.006* (0.003)
Upset Win	-0.002 (0.003)	-0.012*** (0.003)	-0.013*** (0.003)	-0.009* (0.005)	-0.015*** (0.005)	-0.019*** (0.007)
Predicted Win	-0.002 (0.002)	-0.003 (0.002)	0.006** (0.003)	-0.001 (0.003)	-0.006** (0.003)	0.007** (0.003)
Predicted Close	-0.001 (0.002)	-0.000 (0.002)	0.003 (0.003)	-0.004 (0.003)	-0.002 (0.003)	0.003 (0.003)
Predicted Loss	0.002 (0.002)	0.004* (0.003)	0.008*** (0.003)	0.008** (0.004)	0.007** (0.004)	0.010** (0.005)
<i>Panel B. Non-perishable Food</i>						
Upset Loss	0.003 (0.003)	0.007** (0.003)	0.000 (0.004)	0.006* (0.004)	0.013*** (0.004)	0.005 (0.004)
Close Loss	0.004* (0.002)	0.003 (0.002)	0.002 (0.003)	0.007** (0.003)	0.009*** (0.003)	0.004 (0.004)
Upset Win	-0.004 (0.003)	-0.013*** (0.003)	-0.014*** (0.004)	-0.010 (0.007)	-0.019*** (0.007)	-0.026*** (0.008)
Predicted Win	-0.007** (0.003)	-0.006* (0.003)	0.005 (0.004)	-0.007** (0.003)	-0.008** (0.003)	0.007* (0.004)
Predicted Close	-0.005* (0.003)	-0.002 (0.003)	0.001 (0.003)	-0.008** (0.003)	-0.001 (0.003)	0.002 (0.004)
Predicted Loss	-0.002 (0.003)	0.005 (0.003)	0.007* (0.004)	0.006 (0.004)	0.013*** (0.005)	0.016*** (0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. Perishable food includes dairy, deli, fresh produce, and packaged meat departments. Nonperishable food includes dry grocery and frozen foods departments. These categorizations are based on product hierarchy structured by the NielsenIQ Company. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B8: Emotional Shocks and Household Total Spending - Different Cutoffs

<i>Dependent Var:</i>	All Games			Home Games		
<i>Ln(spending on...)</i>	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.004 (0.003)	0.012*** (0.003)	-0.001 (0.003)	0.010*** (0.004)	0.020*** (0.004)	0.004 (0.004)
Close Loss	0.000 (0.004)	-0.007* (0.004)	0.006 (0.004)	0.012** (0.006)	0.012** (0.006)	0.014** (0.006)
Upset Win	-0.006** (0.003)	-0.011*** (0.003)	-0.006* (0.003)	-0.011* (0.005)	-0.006 (0.006)	-0.012* (0.006)
Predicted Win	-0.009*** (0.003)	-0.007* (0.004)	0.007* (0.004)	-0.009** (0.004)	-0.009** (0.004)	0.008* (0.004)
Predicted Close	-0.005 (0.004)	0.010** (0.004)	-0.005 (0.004)	-0.011** (0.005)	0.007 (0.005)	-0.004 (0.005)
Predicted Loss	-0.001 (0.004)	0.008** (0.004)	0.008* (0.004)	0.004 (0.005)	0.008 (0.005)	0.012** (0.005)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.130	0.158	0.207	0.130	0.157	0.206
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -3 or less, *Predicted Close* indicates a point spread between -3 and 3 (exclusive), and *Predicted Loss* indicates a spread of +3 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for the household's marital status, childbearing, race, and internet access, as well as the household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B9: Emotional Shocks and Household Total Spending - Different Fixed Effects

<i>Dependent Var:</i>	All Games			Home Games		
<i>Ln(spending on...)</i>	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.012*** (0.004)	0.012*** (0.004)	0.004 (0.004)	0.012** (0.005)	0.011** (0.005)	0.006 (0.005)
Close Loss	0.002 (0.003)	-0.000 (0.003)	0.000 (0.003)	0.006 (0.004)	0.006 (0.004)	0.006 (0.004)
Upset Win	0.002 (0.004)	-0.004 (0.004)	-0.008* (0.005)	0.010 (0.009)	-0.014 (0.009)	-0.016* (0.010)
Predicted Win	-0.017*** (0.004)	-0.003 (0.004)	0.006 (0.004)	-0.012*** (0.004)	-0.002 (0.004)	0.009** (0.005)
Predicted Close	-0.007** (0.004)	0.005 (0.004)	0.002 (0.004)	-0.009** (0.004)	0.004 (0.004)	0.002 (0.005)
Predicted Loss	-0.004 (0.004)	0.007* (0.004)	0.006 (0.004)	0.001 (0.006)	0.016** (0.006)	0.014** (0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season - week FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj.R ²	0.131	0.160	0.207	0.131	0.159	0.207
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** p<0.01, ** p<0.05, * p<0.1.

Table B10: Emotional Shocks and Household Total Spending - Control for Last Week Spending

<i>Dependent Var:</i>	All Games			Home Games		
<i>Ln(spending on...)</i>	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.004 (0.004)	0.010*** (0.004)	-0.001 (0.004)	0.008* (0.004)	0.016*** (0.005)	0.003 (0.005)
Close Loss	0.004 (0.002)	0.002 (0.003)	0.003 (0.003)	0.011*** (0.004)	0.010*** (0.004)	0.008* (0.004)
Upset Win	-0.003 (0.004)	-0.014*** (0.004)	-0.012*** (0.004)	-0.010 (0.008)	-0.021** (0.008)	-0.023** (0.009)
Predicted Win	-0.010*** (0.004)	-0.007* (0.004)	0.007* (0.004)	-0.010** (0.004)	-0.010** (0.004)	0.008** (0.004)
Predicted Close	-0.006* (0.003)	0.002 (0.004)	0.002 (0.004)	-0.010*** (0.004)	0.002 (0.004)	0.002 (0.004)
Predicted Loss	-0.002 (0.004)	0.008** (0.004)	0.007* (0.004)	0.007 (0.005)	0.015*** (0.006)	0.014** (0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.139	0.157	0.209	0.138	0.156	0.209
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B11: Emotional Shocks and Household Total Spending - Clustering at HH and season level

<i>Dependent Var:</i> <i>Ln(spending on...)</i>	All Games			Home Games		
	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.005 (0.004)	0.010*** (0.004)	-0.001 (0.004)	0.009** (0.004)	0.016*** (0.005)	0.003 (0.005)
Close Loss	0.004* (0.003)	0.003 (0.003)	0.002 (0.003)	0.011*** (0.004)	0.010*** (0.004)	0.007* (0.004)
Upset Win	-0.002 (0.004)	-0.014*** (0.004)	-0.012*** (0.004)	-0.007 (0.008)	-0.022*** (0.008)	-0.025*** (0.009)
Predicted Win	-0.010*** (0.004)	-0.007* (0.004)	0.007* (0.004)	-0.010** (0.004)	-0.010** (0.004)	0.008** (0.004)
Predicted Close	-0.006* (0.004)	0.002 (0.004)	0.002 (0.004)	-0.010*** (0.004)	0.003 (0.004)	0.002 (0.004)
Predicted Loss	-0.003 (0.004)	0.008** (0.004)	0.008* (0.004)	0.007 (0.005)	0.016*** (0.006)	0.015** (0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.130	0.158	0.207	0.130	0.157	0.206
Observations	5,344,585	5,344,585	5,344,585	2,848,705	2,848,705	2,848,705

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household and game season level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B12: Emotional Shocks and Household Total Spending - Exclude HHs with over a month of zero spending

<i>Dependent Var:</i> <i>Ln(spending on...)</i>	All Games			Home Games		
	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.005 (0.004)	0.011*** (0.004)	0.001 (0.004)	0.010** (0.005)	0.017*** (0.005)	0.005 (0.005)
Close Loss	0.006** (0.003)	0.003 (0.003)	0.002 (0.003)	0.013*** (0.004)	0.012*** (0.004)	0.007* (0.004)
Upset Win	-0.004 (0.004)	-0.016*** (0.004)	-0.013*** (0.005)	-0.006 (0.009)	-0.026*** (0.009)	-0.028*** (0.009)
Predicted Win	-0.013*** (0.004)	-0.008** (0.004)	0.003 (0.004)	-0.011*** (0.004)	-0.011** (0.004)	0.005 (0.004)
Predicted Close	-0.008** (0.004)	0.002 (0.004)	0.000 (0.004)	-0.013*** (0.004)	0.002 (0.004)	0.001 (0.005)
Predicted Loss	-0.003 (0.004)	0.012*** (0.004)	0.010** (0.004)	0.005 (0.006)	0.020*** (0.006)	0.018*** (0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.137	0.155	0.186	0.138	0.154	0.186
Observations	4,834,408	4,834,408	4,834,408	2,577,009	2,577,009	2,577,009

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B13: Emotional Shocks and Household Total Spending - Different Methods for Last Week's Shopping Data

<i>Dependent Var:</i> <i>Ln(spending on...)</i>	All Games			Home Games		
	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.009** (0.003)	0.008** (0.004)	-0.001 (0.004)	0.011*** (0.004)	0.014*** (0.004)	0.000 (0.005)
Close Loss	0.003 (0.002)	0.000 (0.003)	0.000 (0.003)	0.011*** (0.004)	0.007* (0.004)	0.004 (0.004)
Upset Win	0.002 (0.004)	-0.018*** (0.004)	-0.011*** (0.004)	-0.008 (0.008)	-0.023*** (0.008)	-0.009 (0.009)
Predicted Win	-0.012*** (0.004)	-0.006 (0.004)	0.006 (0.004)	-0.009** (0.004)	-0.010*** (0.004)	0.009** (0.004)
Predicted Close	-0.008** (0.004)	0.000 (0.004)	0.004 (0.004)	-0.012*** (0.004)	0.002 (0.004)	0.003 (0.004)
Predicted Loss	-0.004 (0.004)	0.007* (0.004)	0.007 (0.004)	0.007 (0.005)	0.019*** (0.006)	0.007 (0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.139	0.156	0.186	0.140	0.155	0.186
Observations	5,746,053	5,746,053	5,746,053	3,049,896	3,049,896	3,049,896

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for household's marital status, childbearing, race, and internet access, as well as household's age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B14: Emotional Shocks and Household Total Spending - Team Fans Defined at the County Level

<i>Dependent Var:</i> <i>Ln(spending on...)</i>	All Games			Home Games		
	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss	0.012** (0.006)	0.012** (0.006)	0.005 (0.006)	0.015** (0.007)	0.011 (0.007)	0.009 (0.008)
Close Loss	-0.001 (0.004)	-0.002 (0.004)	0.000 (0.004)	0.010* (0.006)	0.007 (0.006)	0.000 (0.007)
Upset Win	-0.004 (0.006)	-0.024*** (0.007)	-0.011 (0.007)	-0.017 (0.013)	-0.031** (0.013)	-0.021 (0.014)
Predicted Win	-0.021*** (0.006)	-0.005 (0.006)	0.008 (0.006)	-0.020*** (0.006)	-0.008 (0.006)	0.005 (0.007)
Predicted Close	-0.010* (0.006)	0.005 (0.006)	0.008 (0.006)	-0.017*** (0.006)	0.008 (0.006)	0.013* (0.007)
Predicted Loss	-0.007 (0.006)	0.010 (0.006)	0.008 (0.007)	0.010 (0.009)	0.024*** (0.009)	0.015 (0.010)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.135	0.160	0.209	0.134	0.159	0.209
Observations	2,123,409	2,123,409	2,123,409	1,136,867	1,136,867	1,136,867

Notes: The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for households' marital status, childbearing, race, and internet access, as well as households' age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B15: Emotional Shocks and Household Total Spending - Falsification Test

<i>Dependent Var:</i>	All Games			Home Games		
<i>Ln(spending on...)</i>	Sunday	MonToTue	MonToSat	Sunday	MonToTue	MonToSat
Upset Loss-following month	-0.000 (0.004)	-0.004 (0.004)	-0.009* (0.005)	-0.003 (0.005)	0.006 (0.006)	-0.002 (0.006)
Close Loss-following month	-0.001 (0.003)	0.000 (0.003)	0.003 (0.003)	0.002 (0.004)	0.001 (0.004)	0.004 (0.004)
Upset Win-following month	0.000 (0.004)	-0.001 (0.004)	0.001 (0.005)	0.003 (0.006)	-0.001 (0.006)	-0.007 (0.007)
Predicted Win-following month	0.010** (0.004)	-0.006 (0.004)	0.000 (0.005)	0.021*** (0.006)	-0.008 (0.006)	0.001 (0.007)
Predict Close-following month	0.005 (0.004)	-0.004 (0.004)	-0.005 (0.004)	0.013** (0.006)	-0.007 (0.006)	-0.005 (0.006)
Predict Loss-following month	0.012*** (0.004)	-0.004 (0.004)	-0.002 (0.005)	0.020*** (0.006)	-0.003 (0.006)	0.002 (0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Season & Week of season FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.133	0.161	0.205	0.133	0.160	0.205
Observations	4,419,981	4,419,981	4,419,981	2,397,646	2,397,646	2,397,646

Notes: In this table, we investigated whether households' shopping decisions in a given week are impacted by the game results of the following month as a falsification test. The sample is restricted to expenditures following Sunday games or bye weeks during the regular NFL football season from 2004 to 2019. *Predicted Win* indicates a point spread of -4 or less, *Predicted Close* indicates a point spread between -4 and 4 (exclusive), and *Predicted Loss* indicates a spread of +4 or more. Standard errors, which are clustered at the household level, are reported in parentheses. Controls include indicators for households' marital status, childbearing, race, and internet access, as well as households' age, education, household size, and real income. In addition, controls include the logarithm of total spending on the last Saturday (i.e., the day before the Sunday game) and an indicator for holiday weeks (i.e., Christmas Eve, Christmas Day, New Year's Eve, New Year's Day, Halloween, Thanksgiving, Labor Day, Columbus Day, and Veterans Day weekends). *Bye Weeks* is the omitted category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.